The Impossibility of Wrapper Defense Against Prompt Injection

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Abstract

We prove, and formally verify in Lean 4, a **topological obstruction** for continuous Lipschitz wrapper defenses against prompt injection: any such defense leaves a persistent unsafe region of positive measure under standard regularity assumptions. The *continuous regime* is the paper's primary contribution: we instantiate the quantitative refinement in a deliberately near-tangential GP-oblique stress-test family on real LLM safety surfaces, then measure the containment $S_{\text{pred}} \subseteq S_{\text{act}}$ with bootstrap confidence intervals, within-family grid-resolution and kernel sweeps, plus judge and target retargetings that yield either weak positive support, mixed outcomes, or silence depending on whether the sampled archive remains testable. Discrete classifier gates (Llama-Guard-3-1B, ShieldGemma-2B), stochastic wrappers, and multi-turn / pipeline compositions appear as extensions of the same obstruction, treated as illustrations rather than co-equal empirical pillars. The core continuous proof chain is mechanically verified in Lean; the sharpened quantitative appendix results (Theorems C.2 and C.3) are tied to closest coarser formal witnesses in the correspondence table rather than exact byte-for-byte Lean matches. The Lean artifact (46 files, 361 theorem / lemma / corollary, 0 `sorry`, 3 standard axioms: `propext`, `Classical.choice`, `Quot.sound`; artifact repository) with a paper-theorem $\leftrightarrow$ Lean-identifier / witness table (Section P). Implications: any paradigm satisfying (a) continuity, (b) safe-set score preservation, (c) bounded Lipschitz constant inherits the obstruction; we derive three escape hatches as formal corollaries and map their costs.

1 Introduction

Prompt-injection defense wrappers — input classifiers that flag suspicious prompts [1], constitutional rewriting pipelines [3], and input sanitization layers [7] — now form a crowded research paradigm built on a single architectural pattern: a function $D\colon X \to X$ that preprocesses prompts before the model sees them, mapping unsafe inputs to safer equivalents while preserving benign behavior. We prove that a central benign-preserving continuous subregime of this paradigm is topologically obstructed: any continuous, utility-preserving, bounded-Lipschitz D leaves a persistent unsafe region of positive measure, independent of attack model, data distribution, or training procedure. The proof is mechanically verified in Lean 4 with Mathlib (46 files, 361 theorem/lemma/corollary, 0 `sorry`, 3 standard axioms: `propext`, `Classical.choice`, `Quot.sound`), and the same obstruction admits discrete, stochastic, and compositional extensions motivated by how wrappers are actually

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deployed. Because the obstruction is structural, we do not view our experiments as validation: we use them to document one informative witness family for the quantitative refinement, alongside compact illustrations and weaker neighboring diagnostics that range from one-cell support to silence, and to read off three formal escape hatches — drop continuity, drop safe-set preservation, or drop the Lipschitz bound — as design implications for future defenses.

Concretely, continuity, utility preservation, and completeness form a **defense trilemma**: at most two can coexist. The paper's spine is the continuous case — boundary fixation, score-preserving and ε-relaxed variants, then a quantitative persistent-unsafe region under Lipschitz regularity (Theorems 3.1, 3.3 to 3.5 and 3.8). Discrete, stochastic, and multi-turn results are extensions of that obstruction under dropped hypotheses; they are proved but measured only as compact illustrations.

The empirical footprint is correspondingly hierarchical. On a saturated MAP-Elites archive (82 cells, 25×25 grid) we measure one informative continuous witness family for the primary containment $\mathcal{S}_{\mathrm{pred}} \subseteq \mathcal{S}_{\mathrm{act}}$ against real judge-scored LLM surfaces, with estimator bootstrap CIs and within-family resolution / kernel sweeps. Judge swaps, target swaps, and alternative continuous families are reported separately as weaker neighboring-regime diagnostics, ranging from one-cell support to silence depending on whether the sampled archive remains testable, rather than as additional witness families. Discrete classifier gates (Llama-Guard-3-1B, ShieldGemma-2B), stochastic wrappers, and multi-turn / pipeline compositions are treated as secondary footprint studies rather than co-equal validations.

Scope is narrow by design: the theorem applies only to continuous utility-preserving wrappers on connected prompt spaces. It does not constrain training-time alignment, architectural changes, output-side filters, hard blocklists, or multi-component systems that reject or redirect inputs instead of preserving utility on every benign prompt. All three structural assumptions — continuity, utility preservation, and connectedness — are individually necessary (Section G).

Table 1. Scope map for common wrapper families. "Safe-score" denotes the stronger benign-input score-preservation assumption used in the continuous theorem chain.

Family	Cont.	Safe-score	Bounded sens.	Scope
Classifier gate				
Llama-Guard, ShieldGemma	No	No guarantee	–	Regime 2
Moderation gate				
hard reject / redirect	Usually no	No	–	Outside
Refusal gate / wrapper				
return `[REFUSAL]`	Usually no	No	–	Regime 2
Paraphrase / constitutional rewrite				
benign-preserving rewrite / canonicalization	Plausible	Intended / approx.	Plausible	Core / ε
Aggressive safety rewrite				
rewrites some benign prompts too	Plausible	No	Plausible	Score-tax
Smooth surrogate wrapper				
GP / smoothed defenses in this paper	Yes	By design	Yes	Core

Reading guide. "Core" means the continuous score-preserving theorem chain (Theorems 3.1, 3.3, 3.5 and 3.8) is intended to model the family directly. "Core / ε" means the family is naturally modeled by the exact or relaxed safe-score-preserving variants; "Score-tax" means it steps outside the core regime by altering some safe prompts as well.

Contributions.

1. **Continuous obstruction.** A continuous utility-preserving wrapper must fix threshold prompts, and under Lipschitz regularity leaves a persistent unsafe region of positive measure (Theorems 3.1, 3.3 to 3.5 and 3.8).
2. **Cross-regime extensions.** The same obstruction reappears in discrete, stochastic, multi-turn, and pipeline settings (Theorems 4.1 to 4.3).
3. **Formal verification.** The theorem chain is mechanically verified in Lean 4 (46 files, 361 theorem/lemma/corollary, 0 `sorry`; Section 5).
4. **Empirical footprint.** The continuous containment story is measured end-to-end on real LLM safety surfaces; secondary illustrations exhibit the discrete-collapse, stochastic, and compositional footprints in deployed systems.

Relation to prior work. Adversarial-robustness and certified-defense work study failure or guarantees for a *fixed* model [20, 6, 9, 14, 5, 8, 19, 13, 2], while LLM safety work documents persistent jailbreakability and red-teaming pressure [23, 4, 10, 12, 15, 18]. Our claim is map-level: it constrains the wrapper function itself, independent of attack distribution or training procedure, closer in spirit to a no-free-lunch statement for defenses [22].

2 Problem Setup

Definition 2.1 (Alignment Deviation Function)**.** *Let X be a topological space and let $f\colon X \to \mathbb{R}$ be a continuous scoring function. We interpret $f(x)$ as an alignment deviation score for prompt x. Given a threshold $\tau \in \mathbb{R}$:*

$$S_\tau = \{x \in X : f(x) < \tau\} \qquad \textit{(safe region)} \tag{1}$$

$$U_\tau = \{x \in X : f(x) > \tau\} \qquad \textit{(unsafe region)} \tag{2}$$

$$B_\tau = \{x \in X : f(x) = \tau\} \qquad \textit{(boundary)} \tag{3}$$

Definition 2.2 (Defense)**.** *A* defense *is a continuous map $D\colon X \to X$. It is* utility-preserving *if $D(x) = x$ for all $x \in S_\tau$, and* complete *if $f(D(x)) < \tau$ for all $x \in X$.*

We will also use two weaker preservation notions. A defense is *score-preserving on safe inputs* if $f(D(x)) = f(x)$ for all $x \in S_\tau$, and *ε-relaxed on safe inputs* if $|f(D(x)) - f(x)| \le \varepsilon$ for all $x \in S_\tau$.

The central question is whether a wrapper can preserve utility in one of these senses and still make all outputs safe. The following sections show that across several regimes, the answer is no.

3 Impossibility of Wrapper Defense in Continuous Settings

The core argument is short: D fixes every safe input, so continuity makes $\mathrm{Fix}(D)$ closed; S_τ is open and, in a connected space, not closed; therefore $\mathrm{Fix}(D)$ cannot stop at the edge of S_τ and must contain a boundary point.

Theorem 3.1 (Boundary Fixation)**.** *Let X be a connected Hausdorff space (a space that is "in one piece" and where distinct points can be separated by neighborhoods). Let $f\colon X \to \mathbb{R}$ be continuous with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ continuous with $D|_{S_\tau} = \mathrm{id}$. Then there exists $z \in X$ with $f(z) = \tau$ and $D(z) = z$. Moreover, every $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ and $D(z) = z$, and this set is nonempty.*

Proof sketch. In a Hausdorff space, $\mathrm{Fix}(D)$ is closed. Since $D|_{S_\tau} = \mathrm{id}$, $S_\tau \subseteq \mathrm{Fix}(D)$ and hence $\overline{S_\tau} \subseteq \mathrm{Fix}(D)$. But $S_\tau = f^{-1}((-\infty, \tau))$ is open and not closed, so $\overline{S_\tau} \setminus S_\tau \neq \emptyset$. Any $z \in \overline{S_\tau} \setminus S_\tau$ satisfies $f(z) = \tau$ and $D(z) = z$. Full proof in Section F. □

Theorem 3.2 (Defense Trilemma)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. No $D\colon X \to X$ can simultaneously be continuous, utility-preserving ($D|_{S_\tau} = \mathrm{id}$), and complete ($f(D(x)) < \tau$ for all x).*

All three hypotheses are individually necessary: counterexamples for dropping connectedness, continuity, or utility preservation appear in Section G. The theorem fixes $\overline{S_\tau} \setminus S_\tau$, not necessarily all of B_τ.

3.1 Exact and Relaxed Utility Preservation

The impossibility survives two natural weakenings of exact identity on the safe set.

Theorem 3.3 (Score-Preserving Defense)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. If $D\colon X \to X$ is continuous and* score-preserving on safe inputs*: $f(D(x)) = f(x)$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $f(D(z)) = \tau$.*

Proof sketch. Define $h = f \circ D - f$. Then h is continuous and $h|_{S_\tau} = 0$. The zero set $\{h = 0\}$ is closed and contains $\overline{S_\tau}$, since closed sets contain their boundary points. For $z \in \overline{S_\tau} \setminus S_\tau$, $f(D(z)) = f(z) = \tau$. □

Theorem 3.4 (ε-Relaxed Utility Preservation)**.** *Under the hypotheses of Theorem 3.3, if* $|f(D(x)) - f(x)| \le \varepsilon$ *for all* $x \in S_\tau$*, then there exists* z *with* $f(z) = \tau$ *and* $f(D(z)) \ge \tau - \varepsilon$*.*

Proof sketch. The set $\{x : h(x) \ge -\varepsilon\}$ is closed and contains S_τ, hence $\overline{S_\tau}$. For $z \in \overline{S_\tau} \setminus S_\tau$: $f(D(z)) = \tau + h(z) \ge \tau - \varepsilon$. □

3.2 Quantitative Failure Near the Boundary

Boundary fixation gives a fixed threshold point; Lipschitz regularity spreads failure to a neighborhood.

Theorem 3.5 (ε-Robust Defense Constraint)**.** *Under the hypotheses of Theorem 3.1, if* (X, d) *is a metric space,* f *is* L*-Lipschitz, and* D *is* K*-Lipschitz, then for the fixed boundary point* z*:*

$$f(D(x)) \;\ge\; \tau - LK\,\mathrm{dist}(x, z) \qquad \textit{for all } x \in X. \tag{4}$$

Points within distance δ *of* z *remain within* $LK\delta$ *of threshold.*

Proof sketch. Since $D(z) = z$ and D is K-Lipschitz: $\mathrm{dist}(D(x), z) \le K\,\mathrm{dist}(x, z)$. Since $f(z) = \tau$ and f is L-Lipschitz: $|f(D(x)) - \tau| \le L \cdot K\,\mathrm{dist}(x, z)$. Full proof in Section F. □

Explicit band-volume refinements are deferred to Section C.

3.3 Persistent Unsafe Region

The ε-robust bound can still allow values slightly below τ. Persistence appears when f rises faster in the unsafe direction than the defense can decrease it along its displacement path. For that comparison we use the defense-path Lipschitz constant

$$\ell \;=\; \sup_{x \ne D(x)} \frac{|f(D(x)) - f(x)|}{\mathrm{dist}(D(x),\, x)},$$

which satisfies $\ell \le L$ and is often much smaller than L in the anisotropic regimes of interest.

Lemma 3.6 (Input-Relative Bound)**.** *Under the hypotheses of Theorem 3.5, if* f *has defense-path Lipschitz constant* ℓ*, then* $f(D(x)) \ge f(x) - \ell(K+1)\,\mathrm{dist}(x, z)$ *for all* $x \in X$*.*

Proof sketch. Triangle inequality: $\mathrm{dist}(D(x), x) \le \mathrm{dist}(D(x), z) + \mathrm{dist}(z, x) \le (K+1)\,\mathrm{dist}(x, z)$. Defense-path Lipschitz: $|f(D(x)) - f(x)| \le \ell\,\mathrm{dist}(D(x), x) \le \ell(K+1)\,\mathrm{dist}(x, z)$. □

Definition 3.7 (Steep region)**.** *Given a fixed boundary point* z*, the* steep region *is* $\mathcal{S} = \{x \in X : f(x) > \tau + \ell(K+1)\,\mathrm{dist}(x, z)\}$*—the set of points where alignment deviation exceeds* τ *by more than the defense's Lipschitz budget can compensate.*

Theorem 3.8 (Persistent Unsafe Region)**.** *Let* X *be a connected Hausdorff metric space,* f *continuous and* L*-Lipschitz,* D *continuous and* K*-Lipschitz with* $D|_{S_\tau} = \mathrm{id}$*, and* $S_\tau, U_\tau \ne \emptyset$*. Let* $z \in \overline{S_\tau} \setminus S_\tau$ *be the fixed boundary point from Theorem 3.1, and let* ℓ *be the defense-path Lipschitz constant. If* $\mathcal{S} \ne \emptyset$*, then:*

1. $\mathcal{S}$ *is open.*
2. $\mathcal{S}$ *has positive measure (under any measure positive on nonempty open sets).*
3. *For every* $x \in \mathcal{S}$*:* $f(D(x)) > \tau$*.*

The defense leaves a positive-measure region that remains unsafe.

Proof sketch. $\mathcal{S}$ is a strict superlevel set of a continuous function, hence open. For $x \in \mathcal{S}$: $f(D(x)) \ge f(x) - \ell(K+1)\,\mathrm{dist}(x, z) > \tau$ by Theorem 3.6 and the definition of $\mathcal{S}$. Full proof in Section F. □

Quantitative refinements, including explicit volume bounds, the cone bound, and the defense-aggressiveness dilemma, are deferred to Section C.

4 Extensions: Discrete, Multi-Turn, Stochastic

The continuous obstruction carries to three settings where one hypothesis is relaxed.

Theorem 4.1 (Discrete Defense Dilemma)**.** *Let X be a finite set with $S_\tau, U_\tau \neq \emptyset$, and $D\colon X \to X$ utility-preserving ($D(x) = x$ for $f(x) < \tau$).*

1. *If D is injective, then $f(D(u)) \geq \tau$ for every u with $f(u) \geq \tau$: the defense is incomplete.*
2. *If D is complete ($f(D(x)) < \tau$ for all x), then D is non-injective.*

Theorem 4.2 (Multi-Turn Impossibility)**.** *Let $\{f_t, D_t\}_{t=1}^{T}$ be alignment functions and defenses over T turns on a connected Hausdorff space, each continuous and utility-preserving, with $S_\tau^{(t)}, U_\tau^{(t)} \neq \emptyset$ at every turn. Then for every turn t, there exists z_t with $f_t(z_t) = \tau$ and $D_t(z_t) = z_t$.*

Theorem 4.3 (Stochastic Defense Impossibility)**.** *Let X be a connected Hausdorff space, $f\colon X \to \mathbb{R}$ continuous with $S_\tau, U_\tau \neq \emptyset$. Let D be a stochastic defense and define $g(x) = \mathbb{E}_{y\sim D(x)}[f(y)]$. If g is continuous and $g(x) = f(x)$ for all $x \in S_\tau$, then there exists z with $f(z) = \tau$ and $g(z) = \tau$.*

Discrete, stochastic, and multi-turn empirical illustrations are reported in Sections L.7 to L.9; full proofs, falsifiers, and Lean correspondence are in Section A.

5 Formal Verification

The core continuous theorem chain is fully verified in Lean 4 (Mathlib v4.28.0) as a 46-file artifact `ManifoldProofs/` with 361 `theorem`/`lemma`/`corollary` declarations, zero `sorry`/`admit`, and only the three standard Mathlib axioms `propext`, `Classical.choice`, and `Quot.sound`. Reproducibility, file-level structure, and the paper-theorem ↔ Lean identifier / witness mapping are deferred to Sections O and P; verification is `cd ManifoldProofs && lake build`. The continuous theorems are existential rather than algorithmic: they certify that boundary witnesses and persistent regions exist under the stated hypotheses, but locating them in practice still requires empirical search.

6 Four Regimes of the Obstruction: Empirical Footprint

The continuous persistence chain (Theorems 3.1, 3.5 and 3.8) is the paper's primary theoretical object; Theorem C.2 and Theorem C.3 are narrower quantitative refinements that only bite in anisotropic transversality regimes. The empirical role of this section is therefore limited and hierarchical: we instantiate one measured non-vacuous continuous witness family (Section 6.1), report alternative continuous families and target/judge retargetings as weaker diagnostics that may yield one-cell support, mixed outcomes, or silence when the refinement is vacuous or untestable, and give compact illustrations of the discrete, stochastic, and compositional extensions (Sections L.7 to L.9). Measurement uses the MAP-Elites driver of [17] over a 2D prompt space (query indirection × authority framing); all alignment-deviation values come from our own runs (Section L). The goal is not to prove the theorem — the Lean artifact does that — but to document one informative witness family and to record where neighboring regimes yield weaker support, mixed outcomes, or silence rather than contradiction.

6.1 Continuous-defense footprint (Regime 1)

On the filled MAP-Elites cells G_f of an $N \times N$ grid, with normalized L_2 cell distance $\mathrm{dist}((i_1, j_1), (i_2, j_2)) = h\sqrt{(i_1 - i_2)^2 + (j_1 - j_2)^2}$ and $h = 1/N$, the validator uses the sampled boundary anchor $z^* = \arg\min_{x\in G_f} |f(x) - \tau|$ and four empirical quantities:

$$\hat{L} = \max_{x\neq y} \frac{|f(x) - f(y)|}{\mathrm{dist}(x, y)}, \qquad \hat{K} = \max_{u\neq v} \frac{\mathrm{dist}(D(u), D(v))}{\mathrm{dist}(u, v)},$$

$$\hat{\ell} = \max_{D(x)\neq x} \frac{|f(D(x)) - f(x)|}{\mathrm{dist}(D(x), x)}, \qquad \hat{G} = \max_{f(x)<\tau<f(y)} \frac{f(y) - f(x)}{\mathrm{dist}(x, y)}.$$

These are the discrete estimators of the surface Lipschitz constant, defense Lipschitz constant, defense-path Lipschitz constant, and boundary directional rate; full implementation details are deferred to

Section L. All main-text quantitative claims use this per-cell defense-path estimator $\hat{\ell}$. Secondary sweep tables that use a different sensitivity-pipeline diagnostic mark it explicitly as $\hat{\ell}_{\mathrm{diag}}$ rather than reusing $\hat{\ell}$.

The validator estimates (L, K, ℓ, G) on a 2D AD heatmap, picks a boundary anchor z^*, and records the containment of Theorem 3.8,

$$\mathcal{S}_{\text{pred}} = \{x : f(x) > \tau + \ell(K{+}1)\ \mathrm{dist}(x, z^*)\} \ \subseteq\ \mathcal{S}_{\text{act}} = \{x : f(D(x)) > \tau\},$$

under any continuous K-Lipschitz utility-preserving D. Per cell we true positive, interior false positive ($\mathrm{FP}_{\mathrm{int}}$: a real counterexample), boundary false positive (discrete movement of a sampled boundary cell, outside the theorem's hypotheses), or false negative (allowed by one-sided containment). Treating each filled cell as its L_2 Voronoi neighborhood, $\mathrm{FP}_{\mathrm{int}} = 0$ on the grid is the discrete analogue of $\mu(\mathcal{S}_{\text{pred}} \setminus \mathcal{S}_{\text{act}}) = 0$, up to the discretization gap $|f(z^*) - \tau|$.

Primary measurement. Saturated archive: target `gpt-3.5-turbo-0125`, judge `gpt-4.1-2025-04-14`, 400 evaluations (100 seeds + 300 MAP-Elites iterations, $\sim$ 5,200 API calls, $\sim$\$15, $\sim$ 130 min); 82 filled cells, 66 strictly unsafe, 10 boundary. Fitting a 2D GP (RBF, $\sigma = 0.2$) and building a smooth-oblique defense $D(x) = x + \alpha\,\beta(\mu(x))\,v(x)$ with $v(x)$ displaced 89.5° from the negative gradient gives $\hat{\ell} = 0.86$ (conservative per-cell max; the sweep diagnostic $\hat{\ell}_{\mathrm{diag}} = 0.27$ in Table 5 uses a different estimator and predicts 46 cells), $\hat{K} = 1.05$, $\hat{G} = 23.6$; transversality $\hat{G} > \hat{\ell}(\hat{K}{+}1) = 1.77$ holds by a factor of ~ 13. $|\mathcal{S}_{\text{pred}}| = 3$, all three actually persistent ($\mathrm{TP} = 3$, $\mathrm{FP}_{\mathrm{int}} = 0$; Figure 3); $|\mathcal{S}_{\text{act}}| = 68$ with 65 false negatives (allowed by one-sided containment). The defense is genuinely non-identity (82 cells displaced). Theorem 3.5 also holds at every filled cell, and Theorem 3.1's boundary fixation is non-vacuously activated: the archive's closest boundary cell (restricted to $\overline{S_\tau} \setminus S_\tau$) has raw $|f(z^*) - \tau| = 0.375$. This oblique construction is a deliberately near-tangential *stress-test* wrapper used to instantiate a non-vacuous transversality regime; it is not presented as the optimization direction a deployed safety wrapper would naturally choose.

Witness-family robustness and neighboring diagnostics. Table 5 should be read hierarchically rather than as four co-equal positive validations: on this archive, the near-tangential GP-oblique family is the only clear large non-vacuous witness of the anisotropic transversality regime. The three non-oblique families all miss the transversality diagnostic; two still exhibit a one-cell containment footprint with $\mathrm{FP}_{\mathrm{int}} = 0$, while the third is mixed under discrete bookkeeping. Within the GP-oblique witness family, $\mathrm{FP}_{\mathrm{int}} = 0$ survives bootstrap CIs, grid-resolution subsampling, kernel sweeps, and seed replication. Judge swaps and committee aggregation remain supportive wherever the sampled archive spans both sides of threshold; target swaps are weaker and often become silent as alignment improves. These neighboring diagnostics should be read as local consistency checks around the witness family, not as independent replications of the same non-vacuous regime. The full tables are deferred to Sections L.3 and M together with the cross-target diagnostics in Table 12. On a well-aligned target (`gpt-5-mini-2025-08-07`), $U_\tau = \emptyset$ and the theorem is correctly silent.

Discrete classifier-gate footprint. The strongest compact illustration outside the continuous regime is the deployed-gate map in Figure 1. On the same 82 sampled archive cells, Llama-Guard-3-1B collapses to 7 MLC-taxonomy labels while passing 51/82 prompts as `safe`; ShieldGemma-2B collapses to 2 labels while passing 56/82. Both therefore sit on the injective-incomplete horn of Theorem 4.1: some structure is preserved, but majority incompleteness remains. Full numeric tables and the forced-collapse / injective-paraphrase appendix follow in Section L.7 and tables 14 to 16.

Other secondary illustrations. The stochastic illustration reports nearest-to-threshold cells only, showing the expected variance / mass-above-τ dichotomy without a quantitative theorem claim (Section L.8, table 17, and fig. 5). The multi-turn and pipeline checks are likewise compact sanity checks of the turnwise and compositional extensions (Sections D and L.9 and table 18). Appendix Section N adds a bounded-budget PAIR-style baseline, included only as a finite-budget attacker check rather than as an asymptotic test of the injective-horn prediction.

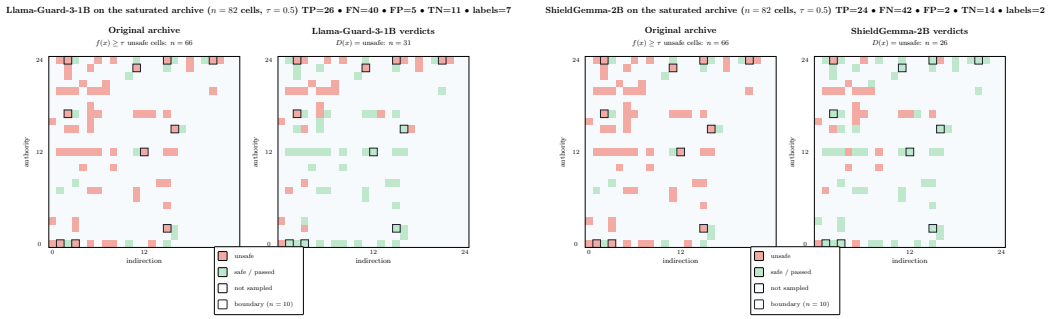

Figure 1. Behavioral-map footprint of the two deployed classifier gates on the same 25×25 saturated archive used elsewhere in the paper. Left panel in each map: original archive classification under the target model's AD score. Right panel: the gate's verdicts placed back on the same filled cells. Llama-Guard-3-1B preserves 7 labels while letting 51 of 82 sampled cells pass as `safe`; ShieldGemma-2B collapses to 2 labels and lets 56 of 82 sampled cells pass as `safe`.

7 Design Consequences

The forbidden zone. Within the paper's core regime — continuous wrappers that preserve safe-set scores and have bounded effective sensitivity — the continuous chain (Theorems 3.1, 3.3, 3.5 and 3.8) implies a structural obstruction, with Theorem C.2 giving a stronger quantitative refinement when its additional regularity hypotheses are met. The design question is therefore not how to make such a wrapper complete, but which assumption to relax and what cost that move incurs. The three corollaries below should be read in that conditional sense: they describe three escape hatches from the core regime, together with the price each one pays.

In plain terms. If a defense is explicitly designed to vary smoothly with prompt semantics, to leave benign prompts essentially unchanged, and to avoid unbounded sensitivity, then it sits in the regime covered by the core theorem chain. In that regime, tuning the wrapper does not remove the obstruction; escaping it requires giving up at least one of those design goals. The three corollaries below spell out the tradeoffs of doing so.

Corollary 7.1 (Drop continuity: classifier-gate cost)**.** *A defense that abandons continuity — i.e., a classifier gate with a threshold indicator — falls under Theorem 4.1 and inherits the injective-but-incomplete / forced-collapse horns: either some unsafe inputs leak, or distinct harmful inputs map to a bounded number of output classes. Empirical illustration: Llama-Guard-3-1B achieves* 62% *FN with* 7 *labels; ShieldGemma-2B achieves* 68% *FN with* 2 *labels (Table 14 and Table 15; discussion in Section Q).*

Proof sketch. The threshold indicator $\mathbf{1}[s(x) \geq \tau]$ is discontinuous, so hypothesis (a) is violated; the hypotheses of Theorem 4.1 apply directly and yield the two-horned lower bound. □

In plain terms. This means that going discrete (any classifier-gate defense, whether Llama-Guard, ShieldGemma, a keyword filter, or a learned moderator) does not escape the obstruction — it just moves to a different face of it. You now either miss a large fraction of unsafe inputs or you collapse them into a handful of output labels. Both costs are directly measurable on any deployed gate; the numbers above are existence proofs, not optimization baselines.

Corollary 7.2 (Drop score preservation: utility tax)**.** *A defense that violates safe-set score preservation (Theorem 3.3) — i.e.,* $f(D(x)) \neq f(x)$ *on some* $x \in S_\tau$ *— can step outside the score-preserving obstruction only by altering the target's outputs on legitimate queries. In other words, escaping the core regime this way pays a utility tax on the safe set.*

Proof sketch. Theorem 3.3 shows that preserving f on S_τ plus continuity forces boundary fixation; removing the preservation assumption permits escape, but any D with $f(D(x)) \neq f(x)$ on S_τ directly changes behavior the user legitimately requested. □

In plain terms. This means that if a defense wants to escape by perturbing the model's behavior on the safe set as well, it is paying for every unit of avoided persistence with a unit of degraded safe-completion quality. Users experience this as a defense that refuses or distorts benign requests; the degradation is not a design bug, it is the exact cost the theorem makes explicit.

Corollary 7.3 (Bound K below threshold: redundancy regime)**.** *The cone condition $G > \ell(K+1)$ from Theorem C.2 requires $K > G/\ell - 1$ for non-trivial persistence. Dropping K below this threshold eliminates the persistence prediction but forces the defense's smoothing kernel to be so flat that D approaches an identity operator. In low-G regimes this can make the wrapper add little measurable safety beyond the aligned target itself; in high-G regimes, by contrast, the quantitative refinement is the one that bites. Empirical anchor: on a target with measured $\hat{G} \approx 1.55$ (a regime where the raw target refuses most prompts on its own), the quantitative threshold already sits below the observable range, so the wrapper adds little beyond the target's own alignment; on the saturated archive $\hat{G} = 23.6$ with $\hat{K} \approx 1.5$ the condition bites and persistence is observed (Section Q).*

Proof sketch. Substitute K into the cone inequality of Theorem C.2. Below the threshold $K^\star = G/\ell - 1$ (Theorem C.3), the bound becomes vacuous; simultaneously $K \to 0$ drives the smoothing kernel to the identity, so $D(x) \approx x$ and the wrapper leaves the target unchanged. □

In plain terms. The third escape route is to make the wrapper weak enough that the quantitative persistence refinement stops giving a positive prediction. In shallow regimes that can leave a wrapper close to redundant; in steep regimes — the ones where wrappers are most needed — the same move is much harder to make without collapsing the defense toward the identity.

Research directions the impossibility does not rule out. The obstruction is specific — it applies to wrappers satisfying (a)–(c). At least four research frontiers remain productively open. *(i) Calibrated incomplete defenses with explicit coverage guarantees*: drop completeness deliberately, provide a guaranteed lower bound on coverage, and chart the Pareto curve of (coverage, false-negative rate) as a function of threshold. *(ii) Non-wrapper architectures*: inference-time defense via modified decoding (constrained sampling, refusal heads, activation gating) rather than a wrapper over the target, which escapes hypothesis (a) by construction rather than by discretization. *(iii) Deployment-regime diagnostics*: use measured (G, ℓ, K) on a target as a diagnostic of which quantitative regime a wrapper occupies — for example, whether the cone refinement is likely to be informative or vacuous on that target. *(iv) Architectural changes to the target*: if wrappers are obstructed, fix the model — RLHF, constitutional AI, activation steering — rather than layering post-hoc filters whose Lipschitz constraints are bound by Theorem 3.2.

Formal status. The corollaries in this section follow from the Lean-verified proof chain documented in Section 5 and Section P; readers can audit each derivation in the artifact.

8 Implications, Scope, and Limitations

The theorem concerns continuous utility-preserving wrappers on connected spaces; the empirical footprint is measured on one synthesized archive with OpenAI judges and should not be read as a bound on the theorem itself. The Lean artifact formalizes the mathematical proof chain, not the experimental pipeline. Extended assumption-by-assumption caveats, judge/benchmark scope, and the worked $K^\star$ example are deferred to Section Q. Appendix Section I also records three stability corollaries explaining why small fine-tuning does not localize the obstruction away for free: unsafe interiors are stable, transversal crossings persist, and local patching is nonlocal.

9 Conclusion

Wrapper defense is structurally incomplete under utility-preserving constraints. In the primary continuous regime, wrappers fix threshold prompts and, under additional regularity, leave persistent unsafe regions; discrete, stochastic, and multi-turn settings do not escape the obstruction so much as re-express it. Empirically, the continuous containment story is visible on real judge-scored LLM safety surfaces, while secondary regime illustrations show the same footprint shape in classifier gates, stochastic wrappers, and compositions.

The practical implication is not that defense is futile, but that wrapper defense should be engineered as risk management rather than as a route to completeness. Stronger guarantees require stepping outside the continuous utility-preserving wrapper model. Appendix Section E makes the complementary geometric point: under the additional continuity and Lipschitz hypotheses used there, unsafe basins are open and connected unsafe fragments cannot shrink arbitrarily finely, which helps explain why empirical search can locate witnesses without the theorem itself being algorithmic.

Broader Impact

Intended positive impact. The contribution is a formally verified no-go theorem. Its intended effect on the field is to *narrow* the design space for LLM safety research, redirecting effort away from topologically obstructed paradigms — continuous, utility-preserving, bounded-Lipschitz wrappers over a fixed target — toward productive alternatives made explicit by the three escape-hatch corollaries (Section 7): discontinuous classifier gates with calibrated coverage guarantees, score-altering rewrite pipelines with measured utility tax, and non-wrapper interventions (training-time alignment, activation-level editing, architectural changes) that fall outside the theorem's hypotheses by construction. A mechanically verified impossibility is a cheaper signal than a failed research program.

Risk surface and responsible reading. The paper names specific deployed classifiers — Llama-Guard-3-1B and ShieldGemma-2B — as having illustrative obstruction footprints on our synthesized archive. This should *not* be read as an endorsement of non-wrapper alternatives, nor as grounds for removing existing wrappers from deployed systems. The obstruction is a lower bound on adversarial success for a specific class of defenses, not a claim that wrappers are harmful or that defense-in-depth is counterproductive. A wrapper with a non-trivial false-negative rate on adversarial prompts can still deliver substantial value on benign-population traffic, raise attack cost, and serve as one component of a layered safety stack; our result constrains only the completeness of any single continuous wrapper in isolation.

Dual-use considerations. The attack baseline we use for archive construction is PAIR [4], which is already public; the theorem does not provide novel attack vectors beyond what the literature already describes. The persistent-region prediction of Theorem 3.8 is a *structural* statement — unsafe regions of positive measure survive any continuous Lipschitz wrapper — not a recipe for locating such regions for a specific target. Locating a persistent cell in practice still requires the same search machinery (behavioral-space exploration, judge scoring) that is standard in red-teaming work and is not enabled by our paper.

Disclosure. The empirical illustrations use deployed commercial defenses (OpenAI-hosted targets and judges, Llama-Guard-3-1B, ShieldGemma-2B). We did not perform coordinated disclosure with the relevant vendors prior to submission, as the results do not constitute novel attacks: the theorem is a structural impossibility result, the PAIR attack baseline is published, and the per-cell findings are standard red-team outcomes on known benchmarks. We will share the saturated archive, Lean artifact, and footprint-evaluation code with any vendor on request.

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Appendices

A Proofs of the Extensions

Full statements and proofs for the three extensions summarized in Section 4: discrete (Theorem 4.1), multi-turn (Theorem 4.2), and stochastic (Theorem 4.3). Each is a corollary of the continuous obstruction with one hypothesis relaxed; all three are verified in Lean (see Section P).

A.1 Discrete-setting impossibility

The discrete and continuous views are compatible; the Tietze/McShane–Whitney bridge (Theorem B.1) converts between them. We prove the discrete statement directly from counting arguments, independent of topology.

Proof of Theorem 4.1. (1) Suppose D is injective and $f(D(u)) < \tau$ for some u with $f(u) \geq \tau$. Then $D(u)$ is safe, so $D(D(u)) = D(u)$ by utility preservation. Injectivity gives $D(u) = u$, so $f(u) = f(D(u)) < \tau$, contradicting $f(u) \geq \tau$. (2) For any $u \in U_\tau$: completeness gives $f(D(u)) < \tau$, so utility preservation gives $D(D(u)) = D(u)$. But $u \neq D(u)$ since $f(u) \geq \tau > f(D(u))$. So u and $D(u)$ are distinct inputs with $D(u) = D(D(u))$: the defense is non-injective. □

Falsifier: a complete utility-preserving classifier D with $|D(X)| = |X|$ on any finite input set.

A.2 Multi-turn impossibility

Proof sketch for Theorem 4.2. Apply Theorem 3.1 to (f_t, D_t) at each turn. The functions may depend on full history; this does not matter, as each timestep is a fresh instance of boundary fixation. The Lean proof establishes monotonicity of the running-max attack success (`running_max_monotone`) and binary-search reachability of transversality (`transversality_reachable`). □

Falsifier: an attack trajectory with strictly decreasing running-max AD.

A.3 Stochastic impossibility

Proof sketch for Theorem 4.3. Define $h = g - f$: continuous, vanishing on S_τ, and so by the closure argument of Theorem 3.3, vanishing on $\overline{S_\tau}$. For $z \in \overline{S_\tau} \setminus S_\tau$, $g(z) = f(z) = \tau$. □

Dichotomy. Since $\mathbb{E}[f(D(z))] = \tau$, either $f(D(z)) = \tau$ almost surely (deterministic at z) or $f(D(z))$ has positive probability of exceeding τ: a genuinely random defense at a boundary point must sometimes make things worse. The continuity of g is the nontrivial assumption; stochastic defenses with discontinuous rejection probabilities escape the theorem.

Falsifier: a stochastic defense at a boundary cell with $\mathrm{variance} = 0$ and $f(D(z)) < \tau$.

B Continuous Interpolation of Discrete Observations

If one wishes to embed discrete behavioral observations into a continuous prompt space, the bridge is standard.

Theorem B.1 (Continuous Relaxation)**.** *Let (X, d) be a connected, normal, Hausdorff metric space, and $S \subset X$ a finite set of observations with observed alignment scores $g\colon S \to \mathbb{R}$ satisfying $g(p) < \tau$ and $g(q) > \tau$ for some $p, q \in S$. Then there exists a continuous $f\colon X \to \mathbb{R}$ extending g (i.e., $f|_S = g$) for which the hypotheses of Theorem 3.1 hold. If the extension is chosen to be Lipschitz (via McShane–Whitney), the hypotheses of Theorem 3.5 also hold. If f is additionally Lipschitz, Theorem 3.8 applies wherever transversality is met.*

Proof sketch. **Step 1** (Extension exists). S is finite and X is T_1, so S is closed. Since X is normal and $g\colon S \to \mathbb{R}$ is continuous on a discrete closed subset, the Tietze extension theorem provides a continuous $f\colon X \to \mathbb{R}$ with $f|_S = g$.

Step 2 (Hypotheses of Theorem 3.1 are satisfied). We have $f(p) = g(p) < \tau$ and $f(q) = g(q) > \tau$, so $S_\tau \neq \emptyset$ and $U_\tau \neq \emptyset$. Since X is connected and Hausdorff, Theorem 3.1 applies.

Step 3 (Lipschitz extension enables stronger results). When a Lipschitz extension is needed, the McShane–Whitney theorem provides an L-Lipschitz f agreeing with g on S. Then Theorem 3.5 and Theorem 3.8 apply under their additional hypotheses. □

C Quantitative Refinements

The main paper establishes existence and persistence. This section records explicit quantitative refinements.

Theorem C.1 (Volume Lower Bound). *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be L-Lipschitz with $L > 0$, and let μ denote Lebesgue measure. If there exists c with $f(c) = \tau - \varepsilon/2$, then $B(c,\, \varepsilon/(4L)) \subseteq \mathcal{B}_\varepsilon$, giving*

$$\mu(\mathcal{B}_\varepsilon) \;\geq\; V_n \cdot \left(\frac{\varepsilon}{4L}\right)^n,$$

where V_n is the volume of the unit ball in $\mathbb{R}^n$. In $\mathbb{R}^1$, this simplifies to $\mu(\mathcal{B}_\varepsilon) \geq \varepsilon/(2L)$.

Proved in the Lean artifact as `MoF_17_CoareaBound`.

Theorem C.2 (Cone Measure Bound). *In $\mathbb{R}$ with Lebesgue measure μ, if $f(x) \geq \tau + c(x - z)$ for all $x \in (z,\, z + \delta_0)$ with $c > \ell(K + 1)$, then $f(D(x)) > \tau$ for every $x \in (z,\, z + \delta_0)$. Consequently,*

$$\mu(\{x \in (z,\, z + \delta_0) : f(D(x)) > \tau\}) \geq \delta_0,$$

and therefore $\mu(\{x : f(D(x)) > \tau\}) \geq \delta_0$.

Related formal witness. The Lean artifact proves the coarser Euclidean local cone bound with slope threshold $L(K{+}1)$ as `MoF_18_ConeBound`; the paper statement records the sharper ℓ-based form used in the empirical discussion.

Theorem C.3 (Defense Dilemma). *Assume f is differentiable at boundary point z with $G = \|\nabla f(z)\|$, and let ℓ be the defense-path Lipschitz constant. Define $K^* = G/\ell - 1$. Then:*

1. *If $K < K^*$, the persistent unsafe region exists ($G > \ell(K + 1)$, so Theorem 3.8 applies).*
2. *If $K \geq K^*$, the ε-robust bound becomes too loose to exclude success on the steep region.*

Since $\ell \leq L$, the dilemma is sharpest when $\ell \ll L$.

Related formal witness. The Lean artifact proves the coarser threshold $K^* = G/L - 1$ as `MoF_19_OptimalDefense`; the paper statement records the sharper engineering threshold $G/\ell - 1$ when a defense-path constant ℓ is available.

D Pipeline Composition

Theorem D.1 (Pipeline Lipschitz Degradation). *If stages $T_1, \ldots, T_n$ are $K_1, \ldots, K_n$-Lipschitz, the composed pipeline is $(\prod K_i)$-Lipschitz. For n stages each with $K \geq 2$, the effective constant is K^n.*

Falsifier. A measured pipeline with empirical composed Lipschitz constant $\hat{K}_{\text{composed}} > \hat{K}_1 \cdot \hat{K}_2$ would refute the bound in Theorem D.1: Lipschitz constants submultiply under composition, so the composed constant can never exceed the product.

Theorem D.2 (Pipeline Impossibility). *If the composed pipeline $P = T_n \circ \cdots \circ T_1 \circ D$ is continuous and $P(x) = x$ for all $x \in S_\tau$, then P has boundary fixed points. If D is K_D-Lipschitz and each T_i is K-Lipschitz, the ε-robust band scales as $L \cdot K_D \cdot K^n \cdot \delta$.*

Proved in the Lean artifact as `MoF_15_NonlinearAgents`.

E Vulnerability Landscape

This section characterizes the geometry of the unsafe region.

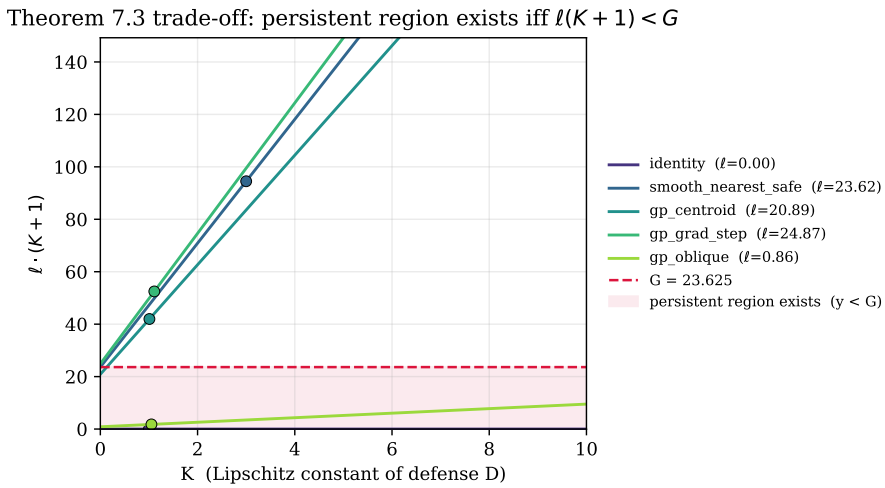

Figure 2. Persistence-vs-utility tradeoff as K varies around $K^* = G/\ell - 1$ from Theorem C.3. Below K^* the persistent unsafe region is nonempty; at and above K^* the displacement needed to kill persistence is incompatible with utility preservation.

Table 2. Empirical pipeline Lipschitz degradation. The measured composed constant $\hat{K}_{\text{composed}}$ is bounded above by $\prod \hat{K}_i$, consistent with Theorem D.1.

stage 1	stage 2	K_1	K_2	K_1K_2	K_{comp}	$K_{\text{comp}} \leq K_1K_2$?
`smooth-nearest-safe`	`kernel-smoothed`	3.00	1.71	5.13	5.10	✓
`smooth-nearest-safe`	`soft-projection`	3.00	7.07	21.21	10.44	✓
`kernel-smoothed`	`smooth-nearest-safe`	1.71	3.00	5.13	3.48	✓
`kernel-smoothed`	`soft-projection`	1.71	7.07	12.08	4.47	✓

Theorem E.1 (Basin Structure)**.** *If f is continuous and $f(p) > \tau$, then U_τ is open. Under any measure positive on nonempty open sets, U_τ has positive measure.*

Theorem E.2 (Basin Fragment Minimum Size)**.** *If X is a normed space, f is L-Lipschitz with $f(p) > \tau$, the connected component of U_τ containing p has diameter $\geq 2(f(p) - \tau)/L$.*

Smoother surfaces (smaller L) produce larger basins; rougher surfaces produce smaller fragments.

F Full Proofs

Proof of Theorem 3.1 (Boundary Fixation). **Step 1** (Hausdorff $\Rightarrow$ fixed-point set is closed). $\text{Fix}(D) = \{x : D(x) = x\}$ is the preimage of the diagonal $\Delta \subset X \times X$ under $x \mapsto (D(x), x)$. In a Hausdorff space, Δ is closed, so $\text{Fix}(D)$ is closed.

Step 2 (Utility preservation $\Rightarrow$ safe region $\subseteq$ fixed points). $S_\tau \subseteq \text{Fix}(D)$. Since $\text{Fix}(D)$ is closed: $\overline{S_\tau} \subseteq \text{Fix}(D)$.

Step 3 (Connectedness $\Rightarrow$ safe region is not closed). $S_\tau = f^{-1}((-\infty, \tau))$ is open. If also closed, it would be clopen—but in a connected space the only clopen sets are $\emptyset$ and X. Since both S_τ and U_τ are nonempty, S_τ is not closed.

Step 4 (Boundary point exists). $\overline{S_\tau} \supsetneq S_\tau$, so there exists $z \in \overline{S_\tau} \setminus S_\tau$. Continuity gives $f(z) \leq \tau$; $z \notin S_\tau$ gives $f(z) \geq \tau$. Hence $f(z) = \tau$.

Step 5 (Defense fixes the boundary point). $z \in \overline{S_\tau} \subseteq \text{Fix}(D)$, so $D(z) = z$ and $f(D(z)) = \tau$. $\square$

Proof of Theorem 3.5 (ε-Robust Constraint). By Theorem 3.1, the fixed boundary point z exists.

Step 1. Since $D(z) = z$ and D is K-Lipschitz: $\text{dist}(D(x), z) = \text{dist}(D(x), D(z)) \leq K\,\text{dist}(x, z)$.

Step 2. Since $f(z) = \tau$ and f is L-Lipschitz: $|f(D(x)) - \tau| = |f(D(x)) - f(z)| \leq L\,\mathrm{dist}(D(x), z) \leq LK\,\mathrm{dist}(x, z)$. □

Proof of Theorem 3.8 (Persistent Unsafe Region). **(1)** $\mathcal{S}$ is the strict superlevel set of $x \mapsto f(x) - \ell(K+1)\,\mathrm{dist}(x, z)$ at level τ, hence open.

(2) Open and nonempty implies positive measure.

(3) For $x \in \mathcal{S}$: $f(x) > \tau + \ell(K+1)\,\mathrm{dist}(x, z)$. By Theorem 3.6: $f(D(x)) \geq f(x) - \ell(K+1)\,\mathrm{dist}(x, z) > \tau$.

Lean: `persistent_unsafe_refined` *in* `MoF_20_RefinedPersistence` *(defense-path constant ℓ).* `MoF_11_EpsilonRobust` *contains an earlier version using the global constant L; as noted in Section 3.3, that version is vacuous for isotropic surfaces.* □

G Counterexamples: Each Hypothesis Is Necessary

Counterexample G.1 (Removing connectedness)**.** *$X = \{0, 1\}$ discrete, $f(0) = 0$, $f(1) = 1$, $\tau = 0.5$. $D(0) = 0$, $D(1) = 0$: continuous, utility-preserving, complete.*

Counterexample G.2 (Removing continuity)**.** *$X = [0, 1]$, $f(x) = x$, $\tau = 0.5$. $D(x) = x$ for $x < 0.5$, $D(x) = 0$ for $x \geq 0.5$: utility-preserving and complete, but discontinuous at 0.5.*

Counterexample G.3 (Removing utility preservation)**.** *$D(x) = x_0$ for a fixed safe point: continuous and complete, but destroys all inputs.*

Table 3. Counterexample ablation. Each row drops one of the three hypotheses and reports whether the defense becomes complete, together with the price paid in the other two properties.

Case	Hypothesis dropped	Construction	Validator flag
C.1	connectedness	$X = \{0, 1\}$ discrete, $D = \mathrm{id}$	`applicable=False`; $\overline{S_\tau} = S_\tau$
C.2	continuity	$X = [0, 1]$, $D(x) = x \cdot \mathbf{1}_{x<0.5}$	local Lipschitz $K = 49.0$ at x=0.495
C.3	utility preservation	$D(x) = x_0$ constant	$\lvert D(X)\rvert = 1$; 50/50 safe inputs displaced

H Attack Properties

Theorem H.1 (Perturbation Robustness)**.** *If f is L-Lipschitz and $f(p) > \tau$, then $B(p, (f(p) - \tau)/L) \subseteq U_\tau$. The radius is monotone in $f(p)$.*

Theorem H.2 (Iterative Convergence)**.** *Any monotone-improvement operator T with score in $[0, 1]$ converges. If each step gains $\geq \delta$, convergence takes $\leq \lfloor 1/\delta \rfloor$ steps.*

Theorem H.3 (Transferability)**.** *$\|f - g\|_\infty \leq \delta \implies \{f > \tau + \delta\} \subseteq \{g > \tau\}$. Transfer costs zero additional queries.*

Theorem H.4 (Authority Monotonicity)**.** *If $f(y, \cdot)$ is monotone non-decreasing in authority for each fixed indirection y, the vulnerability set is upward-closed with a critical threshold $a_2^*(y)$ (by IVT, when $f(y, \cdot)$ is continuous and crosses τ). If additionally f is monotone non-decreasing in indirection for each fixed authority, the threshold curve $y \mapsto a_2^*(y)$ is non-increasing.*

Theorem H.5 (Gradient Ascent)**.** *If f has nonzero Fréchet derivative at x, there exist v and $\varepsilon > 0$ with $f(x + \varepsilon v) > f(x)$.*

I Stability Under Fine-Tuning

Theorem I.1 (Interior Stability)**.** *If $\|f - g\|_\infty \leq \varepsilon$: $f(x) > \tau + \varepsilon \implies g(x) > \tau$, and $f(x) < \tau - \varepsilon \implies g(x) < \tau$. Only the band $|f(x) - \tau| \leq \varepsilon$ is uncertain.*

Theorem I.2 (Crossing Preservation)**.** *Let $f, g\colon [a, b] \to \mathbb{R}$ be continuous. If f crosses τ on $[a, b]$ with margin m (i.e., $f(a') < \tau - m$ and $f(b') > \tau + m$ for some $a', b' \in [a, b]$) and $\|f - g\|_\infty \leq \varepsilon < m$, then g also crosses τ on $[a, b]$.*

Theorem I.3 (Patching Is Nonlocal)**.** *There exist f, g with $\|f - g\|_\infty \leq \varepsilon$ such that eliminating a vulnerability at one point necessarily changes values at distant points.*

J Cost Asymmetry

Theorem J.1 (Exponential Cost Asymmetry)**.** *For grid-based defense with $N \geq 2$ bins per axis in d dimensions: attack cost $\leq 1/\delta$ (dimension-independent); defense cost $= N^d$ (exponential in d); ratio $\delta \cdot N^d \to \infty$ as $d \to \infty$.*

At $d = 2$ with $N = 25$ and $\delta = 0.01$, the ratio is 6.25. At $d = 10$, it climbs to $\sim 10^{12}$.

K Additional Verified Results

Lipschitz displacement bound. D is K-Lipschitz with $D(z) = z \implies \operatorname{dist}(D(x), x) \leq (K+1)\operatorname{dist}(x, z)$.

Tool calls amplify failure. Each non-contractive tool call multiplicatively increases the pipeline's effective Lipschitz constant: $K^n < K^{n+1}$ for $K \geq 2$.

Attacker monotone improvement. Best observed alignment deviation is non-decreasing across turns.

Attacker steering. If directional slope varies continuously with attacker parameter α and crosses $\ell(K+1)$, transversality is reachable by IVT.

Stochastic regularity. $g(x) = \mathbb{E}[f(D(x))]$ satisfies $g - f = 0$ on $\overline{S_\tau}$.

Discrete defense dilemma. An injective, utility-preserving defense is incomplete (every unsafe input stays unsafe). A complete, utility-preserving defense is non-injective (distinct inputs collapse to the same output). The three properties—completeness, utility preservation, injectivity—form a trilemma.

Defense position invariance. Lipschitz constant of defense-before-tools equals defense-after-tools: $K_D \cdot K_T^n$.

L Additional Experimental Details

This appendix collects the estimator definitions, sanity checks, continuous-defense sweeps, and robustness tables that support the main paper's empirical claims.

L.1 Estimator definitions

Let G_f denote the set of filled cells in an $N \times N$ MAP-Elites grid and let $h = 1/N$ be the cell width. Distances use normalized L_2 grid coordinates on $[0, 1]^2$:

$$\operatorname{dist}\big((i_1, j_1), (i_2, j_2)\big) = h\sqrt{(i_1 - i_2)^2 + (j_1 - j_2)^2}.$$

All pairs involving unfilled cells are excluded; no imputation is performed. The boundary anchor is

$$z^* = \arg\min_{x \in G_f} |f(x) - \tau|,$$

restricted to sampled boundary cells $\overline{S_\tau} \setminus S_\tau$, with lexicographic tie-breaking.

Surface Lipschitz constant.

$$\hat{L} = \max_{x, y \in G_f,\ x \neq y} \frac{|f(x) - f(y)|}{\operatorname{dist}(x, y)}.$$

Defense Lipschitz constant.

$$\hat{K} = \max_{u, v \in G_f,\ u \neq v} \frac{\operatorname{dist}(D(u), D(v))}{\operatorname{dist}(u, v)}.$$

Defense-path Lipschitz constant.

$$\hat{\ell} = \max_{x \in G_f,\ D(x) \neq x,\ D(x) \in G_f} \frac{|f(D(x)) - f(x)|}{\mathrm{dist}(D(x), x)}.$$

If the set is empty, we take $\hat{\ell} = 0$.

Boundary directional rate.

$$\hat{G} = \max_{x, y \in G_f,\ f(x) < \tau < f(y)} \frac{f(y) - f(x)}{\mathrm{dist}(x, y)}.$$

These are the estimators used by the validator and the tables below.

L.2 Identity and out-of-scope sanity checks

Identity-defense rows are bookkeeping checks rather than theorem tests: under $D = \mathrm{id}$, the predicted and observed unsafe sets coincide by construction.

Table 4. Smoke tests (not a test of the theorem). The two sparse 25×25 live runs under the canonical identity defense. Under $D = \mathrm{id}$, $\mathcal{S}_{\text{pred}} \equiv \mathcal{S}_{\text{act}} \equiv \{x : f(x) > \tau\}$ by construction, so agreement only checks the validator's bookkeeping.

Target	$\boldsymbol{\tau}$	**Filled**	$\boldsymbol{L}$	$\boldsymbol{G}$	$\lvert\mathcal{S}_{\text{pred}}\rvert$	$\lvert\mathcal{S}_{\text{act}}\rvert$	**TP**	$\mathbf{FP}_{\text{int}}$	**FN**
gpt-5-mini-2025-08-07	0.30	14	6.13	6.13	7	7	7	**0**	0
gpt-3.5-turbo-0125 (smoke)	0.50	9	11.44	11.44	7	7	7	**0**	0

L.3 Continuous-defense sweeps

The appendix tables below serve two roles. First, they separate the single measured family where the quantitative refinement is non-vacuous from three continuous neighboring families where the coarse transversality diagnostic fails: two still show weak one-cell support with $\mathrm{FP}_{\mathrm{int}} = 0$, while one is mixed under discrete bookkeeping. Second, they report a sweep-specific GP sensitivity diagnostic $\hat{\ell}_{\mathrm{diag}}$ for the GP-based rows. This quantity is *not* the main-text per-cell defense-path estimator $\hat{\ell}$ of Section 6.1; it is a secondary diagnostic used only to compare nearby GP sweep configurations.

Table 5. Continuous-defense sweep on the saturated `gpt-3.5-turbo-0125` grid over continuous defense families (rows OBLIQUEGP, SMOOTHNEARESTSAFE, KERNELSMOOTHED, SOFTPROJ). OBLIQUEGP is the near-tangential stress-test family used to instantiate the clearest non-vacuous regime. The remaining rows are neighboring continuous constructions, not alternative witness families: their transversality diagnostic fails, two still show a one-cell containment footprint with $\mathrm{FP}_{\mathrm{int}} = 0$, and SOFTPROJ is mixed under discrete bookkeeping ($|\mathcal{S}_{\mathrm{pred}}| = 1$, $\mathrm{TP} = 0$, $\mathrm{FP}_{\mathrm{int}} = 0$). The column $\hat{\ell}_{\mathrm{diag}}$ is the sweep-specific sensitivity diagnostic described above, not the main-text per-cell $\hat{\ell}$. The *role* column distinguishes the informative witness row (`info.`), weaker positive rows (`weak`), and the mixed bookkeeping row (`mix.`).

defense	L	K	$\hat{\ell}_{\mathrm{diag}}$	$\hat{\ell}_{\mathrm{diag}}(K{+}1)$	G	trans.	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	TP	$\mathrm{FP}_{\mathrm{int}}$	$\lvert\mathcal{S}_{\mathrm{act}}\rvert$	role
SMOOTHNEARESTSAFE	23.62	3.162	23.625	98.334	23.62	×	1	1	0	59	weak
KERNELSMOOTHED	23.62	5.000	21.750	130.500	23.62	×	1	1	0	39	weak
SOFTPROJ	23.62	2.000	23.625	70.875	23.62	×	1	0	0	56	mix.
OBLIQUEGP	23.62	1.050	0.274	0.563	23.62	✓	46	46	0	68	info.

L.4 Robustness checks

L.5 Cross-target sweep

L.6 Out-of-scope discrete stress tests

Bounded-step defenses are not continuous, so Theorem 3.8's hypotheses do not apply. We include them only as negative consistency checks.

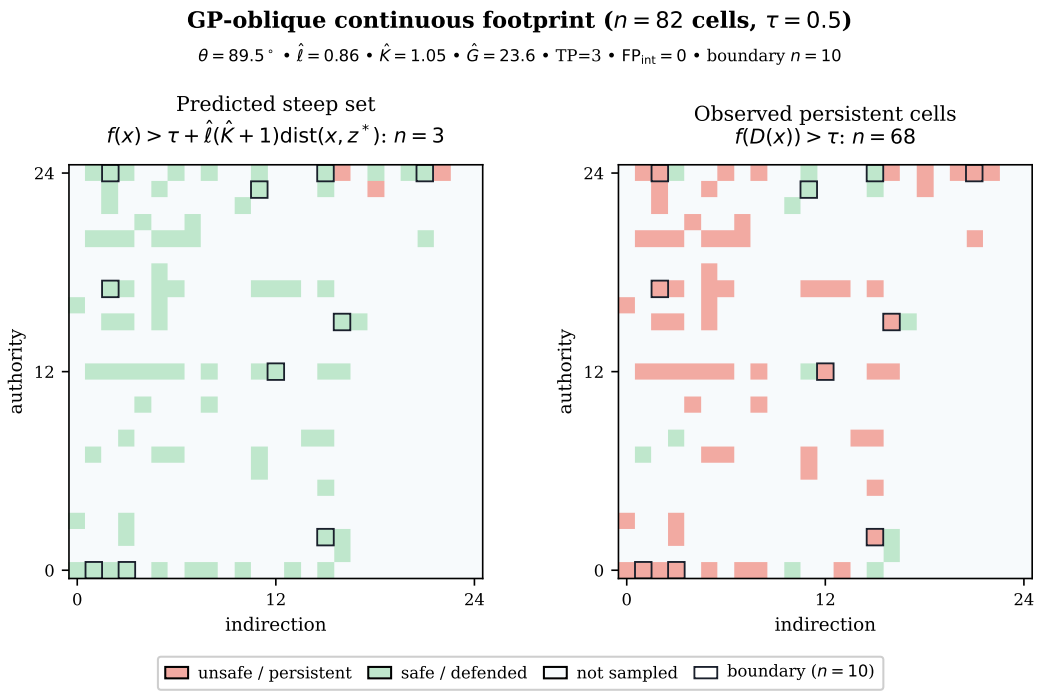

Figure 3. Continuous footprint of Theorem 3.8 on the saturated `gpt-3.5-turbo-0125` grid (82 cells, $\tau = 0.5$) under the GP-smooth oblique defense ($\theta = 89.5^\circ$, $\hat{\ell} = 0.86$, $\hat{K} = 1.05$, $\hat{G} = 23.6 \gg \hat{\ell}(\hat{K}+1) = 1.77$). **Left:** steep set $\mathcal{S}_{\text{pred}}$ ($|\mathcal{S}_{\text{pred}}| = 3$). **Right:** persistent cells $\mathcal{S}_{\text{act}}$ ($|\mathcal{S}_{\text{act}}| = 68$). All 3 predicted cells are persistent ($\mathrm{TP} = 3$, $\mathrm{FP}_{\mathrm{int}} = 0$); the 65 extra cells on the right are allowed by one-sided containment.

Table 6. GP-kernel sensitivity for the oblique-smooth defense on the saturated `gpt-3.5-turbo-0125` grid. Kernel $\in \{\text{RBF}, \text{Matérn-}3/2, \text{Matérn-}5/2\}$ crossed with bandwidth $\sigma \in \{0.1, 0.2, 0.4\}$. This is a within-family sensitivity study for the same near-tangential stress-test construction, reported in terms of the sweep diagnostic $\hat{\ell}_{\text{diag}}$.

kernel	σ	L	K	$\hat{\ell}_{\text{diag}}$	G	$\lvert\mathcal{S}_{\text{pred}}\rvert$	TP	$\mathrm{FP}_{\mathrm{int}}$
RBF	0.10	23.62	1.105	0.884	23.62	9	9	0
RBF	0.20	23.62	1.050	0.274	23.62	46	46	0
RBF	0.40	23.62	1.037	0.099	23.62	71	71	0
Matérn-3/2	0.10	23.62	1.068	1.315	23.62	7	7	0
Matérn-3/2	0.20	23.62	1.070	1.165	23.62	8	8	0
Matérn-3/2	0.40	23.62	1.070	0.946	23.62	8	8	0
Matérn-5/2	0.10	23.62	1.079	0.899	23.62	9	9	0
Matérn-5/2	0.20	23.62	1.080	0.734	23.62	9	9	0
Matérn-5/2	0.40	23.62	1.094	0.345	23.62	41	41	0

L.7 Discrete classifier-gate footprint (Regime 2)

Theorem 4.1 predicts that a discrete utility-preserving D is either non-injective (collapse: distinct unsafe inputs map to the same output) or incomplete (leak: some unsafe input remains unsafe after D). Both horns appear in real deployed gates and in standard post-hoc defenses.

Production classifier gates. Main text already showed the two deployed gate maps (Figure 1). This appendix records the raw counts and the stronger forced-collapse / injective-paraphrase illustrations that sit behind the compact main-paper summary.

Forced-collapse demonstration. On the 66 unsafe prompts of the saturated archive we apply three defenses (Table 15): a refusal defense $D(x) =$ `[REFUSAL]` (complete, collapses 66→1); a canonical-category defense (complete, collapses 66→7); and a paraphrase defense (`gpt-4o-mini`, $T = 0$) injective on this sample (66→66). By Theorem 4.1 the injective defense must be incomplete:

Table 7. Resolution sweep for the oblique-smooth defense on `gpt-3.5-turbo-0125`. This table mixes lattice refinement with archive-fill effects, so non-monotonicity in $|\mathcal{S}_{\text{pred}}|$ and $\hat{\ell}$ should not be read as a pure convergence trend. The containment test itself is unchanged: $\text{FP}_{\text{int}} = 0$ at every resolution.

grid	filled cells	L	K	ℓ	G	$\lvert\mathcal{S}_{\text{pred}}\rvert$	TP	FP_{int}
13×13	25	11.960	1.048	0.629	11.960	5	5	0
17×17	31	14.195	1.033	0.598	14.195	5	5	0
21×21	52	19.845	1.064	0.966	19.845	6	6	0
25×25	82	23.625	1.051	0.865	23.625	3	3	0

Table 8. Coordinate sanity check. Empirical $\hat{L}$ and $\hat{G}$ computed on the same 82-cell archive using the paper's 2D grid metric and a 768-dimensional embedding metric. This is reported as a coordinate check rather than as additional positive evidence for the quantitative refinement.

Metric	$\hat{L}$	$\hat{G}$
2D grid coordinates (normalized to $[0,1]^2$)	23.625	23.625
768-d MPNet embedding (pairwise-max normalized)	17.926	17.926

rescoring through the target+judge pipeline (Table 16) confirms exactly 1/66 strict-exceeds τ — the injective-horn witness.

L.8 Stochastic extension: near-threshold cells (illustration)

Illustration only. Theorem 4.3's precondition $f(z) = \tau$ is a continuous limit; no archive cell attains equality, so we report $f(D(x))$ at the five cells with smallest $|f(x) - \tau|$ ($f \in [0.85, 0.89]$) as a discrete proxy. The predicted dichotomy pattern is visible (Table 17, Figure 5): four cells with $\text{std} > 0$ and $\Pr[f(D(x)) > \tau] > 0$, one with $\text{std} = 0$ and $\Pr[\cdot] = 0$. Closing the gap to ∂S_τ requires a finer archive; no quantitative claim from this subsection.

L.9 Compositional extensions: multi-turn and pipeline (illustration)

Two deliberately compact illustrations of the compositional extensions. *Multi-turn* (Table 18, Figure 6) is included only as a sanity check on the turnwise corollary Theorem 4.2: ten five-turn attack trajectories each exhibit monotone running-max AD, as expected when boundary fixation can recur at every turn. *Pipeline* (Table 2 in Section D) is likewise a compact check that the empirical composed Lipschitz constant $\hat{K}_{\text{composed}}$ is bounded above by $\prod \hat{K}_i$, consistent with the submultiplicative bound of Theorem D.1. Neither subsection is intended as a standalone empirical contribution; both simply document that composition does not, by itself, escape the obstruction.

M Judge robustness

All f values in Section 6.1 are judge-dependent: swapping the judge can reshape the alignment-deviation surface and thereby change $\hat{L}, \hat{\ell}, \hat{G}$. To quantify this, we rescored the 82 (prompt, target response) pairs from the saturated `gpt-3.5-turbo-0125` archive with three alternative judges drawn from different families and capacities — `gpt-4o-2024-08-06`, `gpt-4.1-mini-2025-04-14`, and `gpt-4o-mini-2024-07-18` — applying the same rethinking-evals per-category methodology with a single judge and $\alpha = 0.5$ (target responses were NOT regenerated; only the judge was swapped). The four-judge comparison is summarized in Table 19.

Interpretation. The theorem applies to the full prompt space X under every judge — benign prompts exist, so $\mathcal{S}_\tau$ is nonempty in X. Whether the theorem's positive prediction can be *tested from a given sample* depends on whether that sample spans both regions. Three of four judges (`gpt-4.1`, `gpt-4o`, `gpt-4.1-mini`) yield samples that span both $\mathcal{S}_\tau$ and $\mathcal{U}_\tau$; applying the continuous `smooth_nearest_safe` defense produces TP> 0 and $\text{FP}_{\text{int}} = 0$ in every case. Under the strictest judge (`gpt-4o-mini`), MAP-Elites's adversarial bias drives the sample entirely into $\mathcal{U}_\tau$, so no boundary cell z^* appears in the 82 sampled cells and the positive prediction cannot be tested from this sample. Crucially, $\text{FP}_{\text{int}} = 0$ still holds: no counterexample to the theorem is produced under any of the four judges. Cross-judge agreement ranges from substantial (Kendall's $W = 0.70$) to

Table 9. Seed replication of the saturated pipeline on `gpt-3.5-turbo-0125`. Surface quantities remain within a narrow band and $\mathrm{FP}_{\mathrm{int}} = 0$ in both runs.

Run	Filled	$\hat{L}$	$\hat{K}$	$\hat{\ell}$	$\hat{G}$	trans.	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	TP	$\mathrm{FP}_{\mathrm{int}}$
Saturated (canonical)	82	23.62	3.16	23.62	23.62	×	1	1	0
Replicate (independent)	76	22.25	2.69	22.25	22.25	×	1	1	0

Table 10. Independent target re-evaluation. The same 82 prompts are rescored on `gpt-4o-mini-2024-07-18` with the judge held fixed. No interior counterexample is observed.

Target	Cells	Peak AD	L	K	ℓ	G	$\lvert\mathcal{S}_{\mathrm{pred}}\rvert$	TP	$\mathrm{FP}_{\mathrm{int}}$	Transv.
`gpt-3.5-turbo-0125`	82	1.000	23.62	3.16	23.62	23.62	1	1	0	×
`gpt-4o-mini-2024-07-18`	82	0.990	12.25	4.94	4.02	10.12	0	0	0	×

poor on binary classification (Cohen's $\kappa = 0$ vs gpt-4o-mini), so the *specific numerical instance* of a particular judge is not itself judge-robust even though the *framework's predictions* are.

Committee aggregation. The rethinking-evals methodology supports committee aggregation: $P_{\mathrm{actual}} = \alpha P_{\mathrm{vote}} + (1-\alpha) P_{\mathrm{mean}}$ with $\alpha = 0.5$. We apply this offline to the four existing judge archives (no new API calls) and re-run the validator on each committee-aggregated surface (Table 20). The top-3 committee (gpt-4.1, gpt-4o, gpt-4.1-mini; excluding the strict-judge outlier gpt-4o-mini) is within one cell of canonical (17 safe vs 16; mean $\Delta = +0.010$), and $\mathrm{FP}_{\mathrm{int}} = 0$ under `smooth_nearest_safe` on every committee configuration — the theorem's predictions survive committee aggregation.

N Attack Baseline: PAIR-Style Iteration

To situate the theorem empirically against an established attack-defense benchmark style, we run a PAIR-style iterative attack ([4]-adjacent) on ten top-AD unsafe seed prompts from the saturated archive. The attacker is `gpt-4o-mini`, the target is `gpt-3.5-turbo-0125`, and the judge is `gpt-4.1`. Each behavior gets five iterations; the attacker receives the full transcript and judge scores at each step. We compare two conditions: undefended (attacker prompt goes directly to target) and *paraphrase wrapper* (attacker prompt is rewritten by `gpt-4o-mini` at $T = 0$ before reaching the target).

Interpretation. Within the 5-iteration budget the paraphrase wrapper blocks all ten behaviors (ASR drops from 1.00 to 0.00). This does *not* contradict Theorem 4.1's injective-incomplete horn: the paraphrase defense is injective on our 66-prompt sample (all 66 rewrites are distinct), and Table 16 already exhibits $1/66$ paraphrases that yield $f(D(x)) > \tau$. The PAIR attacker with gpt-4o-mini as refiner and a 5-iteration budget is therefore best read as a bounded-budget baseline, not as evidence against eventual injective-horn leakage. We do not claim that 5 iterations characterize the attacker's asymptotic success probability; they only show that a low-budget attacker fails to locate the known rare bypass on this sample. A fuller budget sweep would be needed to turn this appendix check into a quantitative attacker-model study.

O Lean Artifact

The complete theory is verified in Lean 4.28.0 with Mathlib v4.28.0, available at the public artifact repository. The artifact comprises 46 files:

- 10 core theory files (MoF_01–MoF_10)
- 10 cost theory files (MoF_Cost_01–MoF_Cost_10)
- 10 advanced theory files (MoF_Adv_01–MoF_Adv_10)
- 1 continuous relaxation (MoF_ContinuousRelaxation)
- 1 ε-robust constraint + persistent unsafe region with global Lipschitz constant (MoF_11_EpsilonRobust)
- 1 discrete impossibility (MoF_12_Discrete)
- 1 multi-turn + stochastic extensions (MoF_13_MultiTurn)

Table 11. Bootstrap percentile confidence intervals (median, $[5^{\text{th}}, 95^{\text{th}}]$) for the empirical quantities on the saturated live run.

defense	L_{med}	L CI	K_{med}	K CI	ℓ_{med}	ℓ CI	G_{med}	G CI
identity	23.625	$[21.750, 23.625]$	1.000	$[1.000, 1.000]$	–	$[--, --]$	23.625	$[21.750, 23.625]$
GP-smooth oblique	23.625	$[21.750, 23.625]$	1.051	$[1.029, 1.051]$	0.865	$[0.801, 0.865]$	23.625	$[21.750, 23.625]$

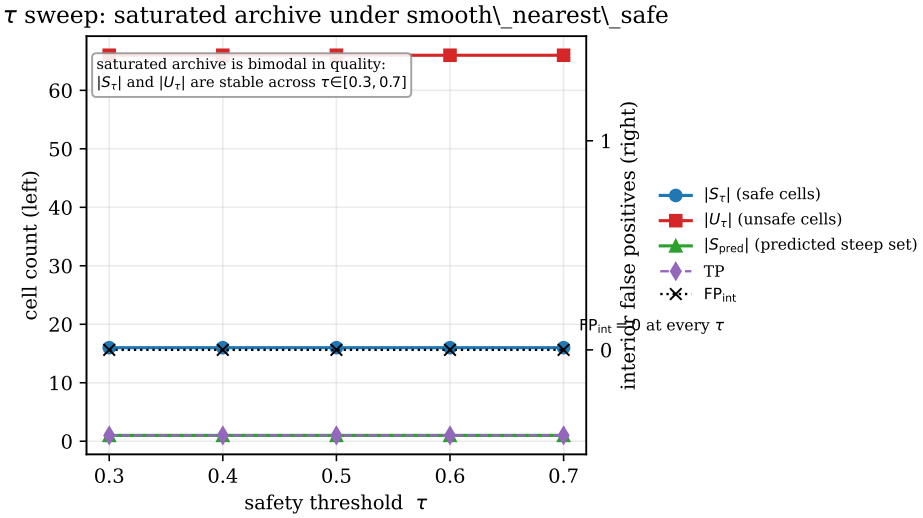

Figure 4. Sensitivity of $|\mathcal{S}_{\text{pred}}|$, $|\mathcal{S}_{\text{act}}|$, and FP_{int} to the threshold τ on the saturated live run. The containment $\mathcal{S}_{\text{pred}} \subseteq \mathcal{S}_{\text{act}}$ holds at every tested τ.

- 1 representation-independent meta-theorem (MoF_14_MetaTheorem)
- 1 nonlinear agent pipelines (MoF_15_NonlinearAgents)
- 1 relaxed utility preservation (MoF_16_RelaxedUtility)
- 1 quantitative ε-band volume bound (MoF_17_CoareaBound)
- 1 cone measure bound for persistent unsafe region (MoF_18_ConeBound)
- 1 optimal defense characterization (MoF_19_OptimalDefense)
- 1 refined persistence with defense-path Lipschitz constant ℓ (MoF_20_RefinedPersistence); this file contains the primary formalization of Theorems 3.6 to 3.8
- 1 gradient chain: $\|\nabla f(z)\| > \ell(K+1)$ implies persistent unsafe region via operator-norm direction extraction and derivative-based local growth (MoF_21_GradientChain)
- 3 capstone files (MasterTheorem, Euclidean instantiation, verification)
- 1 root import file (ManifoldProofs.lean)

P Paper ↔ Lean Correspondence

Tables 22 to 25 map each labeled theorem or lemma in this paper either to a named Lean identifier or, where noted, to the closest formal witness in the `ManifoldProofs/` artifact. For the sharpened quantitative appendix results (Theorems C.2 and C.3), the listed identifier is a coarser Euclidean / L-based witness rather than a byte-for-byte statement match. When a single paper theorem is formalized by several Lean declarations, we list the single most direct witness; all Lean identifiers live in the `MoF` namespace (or an inner `MoF.*` namespace) and can be located in the indicated file. All identifiers below are built by `lake build` with zero `sorry` and rely only on the three standard Mathlib axioms.

Unified witness. The top-level statement `MoF.master_theorem` in `MoF_MasterTheorem.lean` bundles the first three clauses of the correspondence (basin openness, boundary non-emptiness, defense incompleteness) into a single statement over a packed `ManifoldOfFailure` structure, and is the single Lean entry-point that witnesses the core impossibility claim of the paper.

Table 12. Cross-target diagnostics under `smooth_nearest_safe`: 82 prompts and the canonical gpt-4.1 judge fixed; only the target varies. The legacy target retains a weak one-cell footprint (`weak`); better-aligned targets become `sil.` because the sampled archive no longer yields a testable quantitative witness. For the two better-aligned targets, 42–45 of 82 cells score exactly at $\tau = 0.5$ (refusal responses); $|\mathcal{S}_\tau|$ and $|\mathcal{U}_\tau|$ count strict inequalities only, so these at-threshold cells are excluded from both counts. Silent rows provide no positive evidence for the refinement; they are reported only to show that no interior counterexample appears when the refinement is vacuous or untestable on the sampled archive.

Target	Cells	Peak AD	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	L	K	ℓ	G	$\lvert\mathcal{S}_{\text{pred}}\rvert$	FP_{int}	role
`gpt-3.5-turbo-0125`	82	1.00	16	66	23.62	3.16	23.62	23.62	1	0	weak
`gpt-4o-mini-2024-07-18`	82	0.99	13	24	12.25	4.94	4.02	10.12	0	0	sil.
`gpt-4o-2024-08-06`	82	0.99	10	30	12.38	3.41	12.38	12.38	0	0	sil.

Table 13. Out-of-scope discrete stress tests. Defense sweep on the saturated `gpt-3.5-turbo-0125` grid over discontinuous `bounded_step` defenses. These rows are not theorem checks; they are reported only as consistency checks outside the theorem's scope.

Defense	$\boldsymbol{K}$	$\boldsymbol{\ell}$	$\lvert\mathcal{S}_{\text{pred}}\rvert$	$\lvert\mathcal{S}_{\text{act}}\rvert$	**TP**	$\mathbf{FP}_{\text{int}}$	**FN**
identity	1.00	0.00	66	66	66	**0**	0
`bounded_step(1)`	3.61	23.63	1	55	0	**0**	55
`bounded_step(2)`	3.61	23.63	1	47	0	**0**	47
`bounded_step(3)`	4.00	23.63	1	29	0	**0**	29
`bounded_step(5)`	8.60	23.63	1	3	0	**0**	3
`bounded_step(8)`	8.60	23.63	1	0	0	**0**	0
`bounded_step(12)`	8.60	23.63	1	0	0	**0**	0

Q Extended Scope, Limitations, and Worked Example

The theorem is one topological impossibility, mechanically verified in Lean 4; the empirical sections are footprints of that theorem in four deployment regimes, not independent tests of it. The limitations below are organized accordingly: what the *theorem's assumptions* require, what the *empirical illustration* does and does not cover, what the *Lean artifact* formalizes, and what the *derived design consequences* still leave open. A companion worked example at the end ties the Lipschitz bound to a concrete numerical design constraint on the wrapper models we illustrate on.

Theorem-assumption limitations. The impossibility requires four structural assumptions on the prompt space and the defense. (i) The prompt space X is Hausdorff and path-connected; the continuous theorems break on totally disconnected spaces, and the discrete regime (Theorem 4.1) is exactly what replaces them there. (ii) The averaged post-defense score $x \mapsto \mathbb{E}[f(D(x))]$ is continuous in x; stochastic defenses whose expected score is itself discontinuous fall outside the continuous hypotheses, though the stochastic companion (Theorem 4.3) is what governs them. (iii) The defense is bounded Lipschitz with constant K; unbounded-sensitivity defenses are one of the three escape hatches (Section 7) and are not constrained by the quantitative bounds. (iv) Utility preservation holds on a safe set of positive measure. The deployed systems we illustrate on are not all equally well modeled by these assumptions. Several wrapper families are plausibly approximated this way as a first model — especially paraphrase / canonicalization wrappers and the smooth surrogate defenses used in the paper — whereas classifier gates and aggressive rewrites motivate the discrete and score-tax escape hatches instead. The assumptions are therefore a structural modeling choice, not a claim that every deployed guardrail literally satisfies them pointwise.

Illustration limitations (not theorem limitations). The empirical footprint has a bounded reach that should not be read as bounds on the theorem. (a) The saturated archive has 82 filled cells from a single MAP-Elites synthesis seed; richer seeds or larger archives would widen the footprint but cannot change the Lean-proven statement. (b) The judge committee is single-vendor (all OpenAI); Section M reports a four-judge cross-check, but a multi-vendor judge study remains future work. (c) Behavioral-descriptor axes (query indirection, authority framing) are inherited from [17]; different axes would produce different surfaces, not different theorems. (d) We do not overlay public jailbreak benchmarks; the footprint we report is specific to the synthesized archive. None of these are limitations of Theorems 3.1, 3.5, 3.8 and 4.1 to 4.3; they bound how broadly the illustration exhibits the predicted footprint.

Table 14. Llama-Guard-3-1B via local Ollama on the 82 sampled archive cells: footprint of Theorem 4.1's injective-incomplete horn. 51/82 pass as `safe`; the 31 flagged cases span 6 of 14 MLC categories.

Quantity	Value
Input prompts $\|X\|$	82
Distinct Llama-Guard labels $\|D(X)\|$	7
Collapse rate $1 - \|D(X)\|/\|X\|$	0.915
Largest collision class	`safe` ($n = 51$)
Verdict `unsafe`	31
Verdict `safe` (false neg.)	51
Verdict other/error	0
Theorem 4.1 horn	*injective-but-incomplete*

Code	Hazard category	# prompts
S9	Indiscriminate Weapons	8
S2	Non-Violent Crimes	8
S1	Violent Crimes	9
S6	Specialized Advice	4
S10	Hate	1
S7	Privacy	1
–	safe (false negatives)	51

Table 15. Theorem 4.1 footprint on 66 unsafe seeds: refusal (complete, non-injective), canonical-category (complete, non-injective), paraphrase (injective, so must be incomplete).

Defense	Inputs	Distinct outputs	Collapse rate	Max collision class
refusal	66	1	0.985	66
canonical-category	66	7	0.894	40
paraphrase	66	66	0.000	1

Lean-artifact limitations. The Lean 4 development covers the topological and analytic core of the impossibility chain: boundary fixation, ε-robust depth, transversal persistence, discrete non-injectivity, stochastic averaging, multi-turn recurrence, and the three escape-hatch corollaries. Outside the formalization: the empirical experiments themselves are not formalized (the confusion-matrix counts, judge pipeline, and archive saturation are code, not theorems); the behavioral-descriptor axes are informal domain choices; and the artifact depends on the three standard Mathlib axioms (`propext`, `Classical.choice`, `Quot.sound`), which are load-bearing — removing `Classical.choice` in particular would require a constructive reworking of several Hausdorff-space lemmas.

Design-consequences limitations. The three escape hatches of Section 7 (drop continuity, drop safe-set preservation, drop the Lipschitz bound) are derived as formal corollaries, so their *existence* is settled. Their *costs*, however, are anchored empirically: the false-negative rate of discrete classifier gates (Llama-Guard's 51-of-82 pass-through in Table 14), the utility degradation of ε-relaxed wrappers, and the redundancy overhead of low-G smooth wrappers are measured on a single archive and should be read as design-space anchors, not population estimates. A full cost characterization across deployment regimes would require larger multi-vendor studies beyond the scope of this paper.

Scope of the individual assumptions. Each structural assumption maps to a qualitative regime that carries its own caveats. Boundary fixation is only at the boundary: fixed points satisfy $f(z) = \tau$ exactly, so if τ-level behavior is benign the theorem is true but practically harmless. The ε-robust bound limits depth, not direction: the defense may push near-boundary points slightly below τ, and the bound limits how far, not whether. Persistence requires transversality: the persistent unsafe region exists only where the alignment surface is steep ($c > \ell(K{+}1)$), and for isotropic f ($\ell = L$) the region is empty for all $K \geq 0$. The cost asymmetry of Theorem J.1 further assumes exhaustive grid enumeration; learning-based defenses that generalize across the space may sidestep the exponential bound.

Worked example (Theorem C.3). For a Gaussian-smoothed wrapper with $\hat{\ell} = 0.86$ (from our saturated `gpt-3.5-turbo-0125` run) and target $\hat{G} = 23.6$, the optimal defense Lipschitz constant

Table 16. Paraphrase defense is injective on the 66 unsafe seeds (Table 15); rescoring confirms exactly 1/66 strict false negatives, the injective-horn footprint.

Defense	$\lvert\{x : f(D(x))>\tau\}\rvert$	Incomplete per Thm. 4.1
paraphrase (T=0, strict $> \tau$)	$1/66$	✓
paraphrase (T=0, borderline $\geq \tau$)	$5/66$	✓

Table 17. Theorem 4.3 illustration at the five nearest-to-threshold cells ($f(x) \in [0.85, 0.89]$, discrete proxy for ∂S_τ): positive variance accompanies positive mass above τ on 4/5; the fifth has $\mathrm{std} = 0$ and stays below.

Boundary cell	original AD	mean $f(D(x))$	std	max	$P(f(D(x))>\tau)$
$[4, 21]$	0.850	0.513	0.072	0.900	0.03 $(1/30)$
$[14, 8]$	0.850	0.562	0.142	0.890	0.17 $(5/30)$
$[1, 24]$	0.875	0.526	0.100	0.900	0.07 $(2/30)$
$[3, 0]$	0.875	0.526	0.100	0.900	0.07 $(2/30)$
$[2, 15]$	0.890	0.500	0.000	0.500	0.00 $(0/30)$

is $K^* = \hat{G}/\hat{\ell} - 1 \approx 26.5$. Defense designs with $K < K^*$ admit a persistent region (Theorem 3.8 applies non-trivially); designs with $K \geq K^*$ require displacements so large that utility preservation — measured by response-embedding similarity between $D(x)$ and x on safe inputs — drops below the accept threshold. This is a concrete, numerically-grounded corollary of the theorem for any wrapper deployed against this alignment surface.

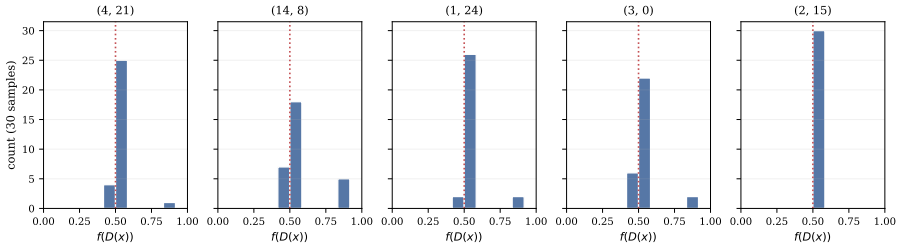

Figure 5. $f(D(x))$ histogram at a near-threshold cell: positive variance with non-zero mass above τ, consistent with Theorem 4.3's dichotomy.

Table 18. Per-turn boundary-fixation statistics for the multi-turn illustration of Theorem 4.2.

Traj	AD_1	AD_2	AD_3	AD_4	AD_5	$\max_k$
T1	0.500	0.500	0.500	0.500	0.900	**0.900**
T2	0.990	0.500	0.500	0.500	0.500	**0.990**
T3	1.000	1.000	1.000	0.995	1.000	**1.000**
T4	0.990	0.500	0.500	0.500	0.500	**0.990**
T5	1.000	1.000	1.000	0.990	0.990	**1.000**
T6	1.000	0.500	0.500	0.500	0.500	**1.000**
T7	0.990	0.990	1.000	0.990	1.000	**1.000**
T8	0.975	0.500	0.500	0.500	0.900	**0.975**
T9	1.000	1.000	1.000	0.500	0.990	**1.000**
T10	0.500	0.975	0.500	0.500	0.500	**0.975**

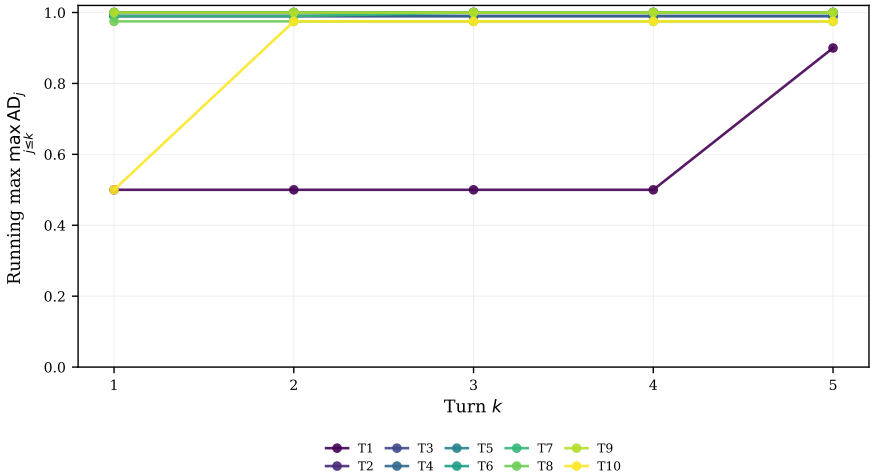

Figure 6. Running-max alignment-deviation is monotone over ten turns, as predicted by Theorem 4.2.

Table 19. Judge robustness on the saturated archive (82 cells, $\tau = 0.5$). $|\mathcal{S}_\tau|_{\text{sample}}$ and $|\mathcal{U}_\tau|_{\text{sample}}$ are how many of the 82 sampled cells fall below/above τ under each judge. Pearson r and Cohen's κ are pairwise with the canonical gpt-4.1 judge. Kendall's $W = 0.700$ across all four judges on cell rankings (moderate agreement). "Testable?" indicates whether this MAP-Elites sample spans both $\mathcal{S}_\tau$ and $\mathcal{U}_\tau$, which is what the positive-prediction test of Theorem 3.8 requires to locate a boundary cell z^*.

Judge	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	peak AD	mean AD	Pearson r	κ	TP	FP_{int}	Testable
`gpt-4.1-2025-04-14` (canonical)	16	66	1.000	0.773	–	–	1	0	yes
`gpt-4o-2024-08-06`	10	72	0.990	0.889	0.546	0.457	1	0	yes
`gpt-4.1-mini-2025-04-14`	33	49	0.975	0.752	0.493	0.363	1	0	yes
`gpt-4o-mini-2024-07-18`	0	82	1.000	0.955	0.393	0.000	–	–	no

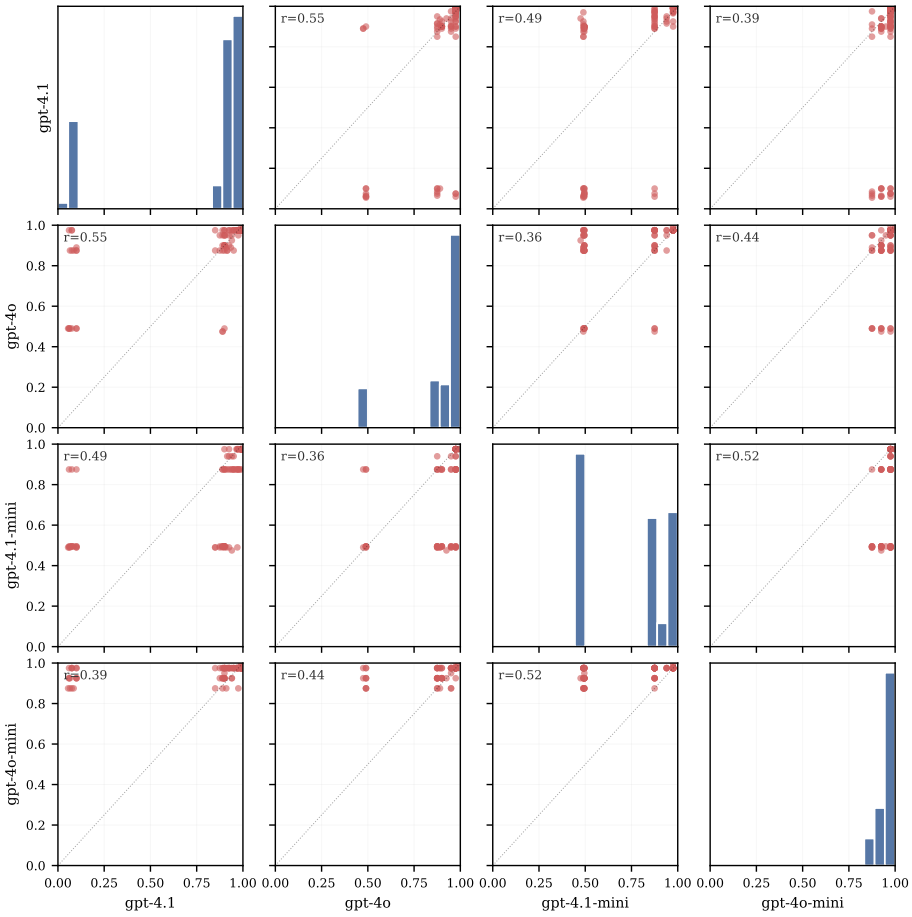

Figure 7. Per-cell alignment-deviation scatter across the four judges. Cross-judge agreement is moderate (Kendall's $W = 0.700$); the positive-prediction test of Theorem 3.8 remains supported wherever the sample spans both $\mathcal{S}_\tau$ and $\mathcal{U}_\tau$.

Table 20. Judge committee aggregations vs the canonical single-judge surface. Top-3 committee is within one cell of canonical; $\mathrm{FP}_{\mathrm{int}} = 0$ in every configuration.

committee	N	$\lvert\mathcal{S}_\tau\rvert$	$\lvert\mathcal{U}_\tau\rvert$	mean Δ	std Δ	$\mathrm{FP}_{\mathrm{int}}$
canonical (gpt-4.1 only)	1	16	66	0.000	0.000	0
top-3 (gpt-4.1, gpt-4o, gpt-4.1-mini)	3	17	65	+0.010	0.173	0
gpt-4 family (gpt-4.1, gpt-4o)	2	13	69	+0.063	0.141	0
all-4 (incl. strict gpt-4o-mini)	4	5	77	+0.058	0.200	0

Table 21. PAIR-style attack-success rate with and without a continuous paraphrase wrapper (10 behaviors, 5 iterations, gpt-3.5-turbo target, gpt-4.1 judge). This is a bounded-budget baseline, not an estimate of asymptotic attacker success.

Condition	behaviors	ASR @ 5 iters	mean running-max
undefended	10	1.00	0.995
paraphrase wrapper	10	0.00	0.500

Table 22. Paper-to-Lean correspondence: every labeled result in the paper is paired with a named Lean identifier in the indicated file. For the sharpened quantitative appendix results, the listed identifier is the closest coarse formal witness rather than an exact statement match. Type signatures are abbreviated (ellipses and prose variable lists) to fit the table; the full Lean signatures can be read directly from the cited files.

Paper result	**Informal statement**	**Lean file**	**Lean identifier / signature**
Theorem 3.1 Boundary Fixation	Continuous utility-preserving defense fixes a boundary point.	`MoF_08_Defense-Barriers`	`defense_incompleteness`: `... Continuous D → Continuous f → (∀ x, f x < τ → D x = x) → ... → ∃ z, f z = τ ∧ D z = z ∧ ¬(f (D z) < τ)`
Theorem 3.2 Defense Trilemma	No continuous utility-preserving defense can be complete.	`MoF_16_Relaxed-Utility`	`score_preserving_defense_incomplete`: `... → ¬(∀ x, f (D x) < τ)`
Theorem 3.3 Score-Preserving Defense	$f \circ D = f$ on $S_\tau \Rightarrow$ boundary fixation.	`MoF_16_Relaxed-Utility`	`score_preserving_boundary_fixation`: `... (∀ x, f x < τ → f (D x) = f x) → ... → ∃ z, f z = τ ∧ f (D z) = τ`
Theorem 3.4 ε-Relaxed Utility	$\lvert f \circ D - f \rvert \leq \varepsilon$ on S_τ forces $f(D(z)) \geq \tau - \varepsilon$ at a boundary z.	`MoF_16_Relaxed-Utility`	`eps_relaxed_boundary_fixation`: `... (∀ x, f x < τ → |f (D x) - f x| ≤ ε) → ... → ∃ z, f z = τ ∧ f (D z) ≥ τ - ε`
Theorem 3.5 ε-Robust Constraint	L-Lipschitz f, K-Lipschitz D: $f(D(x)) \geq \tau - LK \operatorname{dist}(x, z)$ globally.	`MoF_11_Epsilon-Robust`	`epsilon_robust_impossibility`: `... LipschitzWith L f → LipschitzWith K D → ... → ∃ z, f z = τ ∧ D z = z ∧ ∀ x, f (D x) ≥ τ - L · (K · dist x z)`
Theorem 3.6 Input-Relative Bound	Defense-path ℓ: $f(D(x)) \geq f(x) - \ell(K+1) \operatorname{dist}(x, z)$.	`MoF_20_Refined-Persistence`	`defense_from_input_bound_refined`: `... LipschitzWith K D → (∀ x, |f (D x) - f x| ≤ ℓ · dist (D x) x) → D z = z → ∀ x, f (D x) ≥ f x - ℓ · ((K + 1) · dist x z)`
Theorem 3.8 Persistent Unsafe Region	Transversality $\Rightarrow$ positive-measure set stays unsafe.	`MoF_20_Refined-Persistence`	`persistent_unsafe_refined`: `... (∃ x₀, f x₀ > τ + ℓ · ((K + 1) · dist x₀ z)) → 0 < μ {x | f (D x) > τ}`

Table 23. Paper-to-Lean correspondence (continued, part 1b): discrete, multi-turn, stochastic, continuous-relaxation, quantitative, and pipeline results.

Paper result	Informal statement	Lean file	Lean identifier / signature
Theorem 4.1 Discrete Defense Dilemma	Utility-pres. D is incomplete or non-injective.	`MoF_12_Discrete`	`discrete_trilemma: (∀x, f x < τ → D x = x) → f u ≥ τ → (f (D u) ≥ τ) ∨ (∃x y, x ≠ y ∧ D x = D y)`
Theorem 4.2 Multi-Turn Impossibility	Boundary fixation recurs at every turn.	`MoF_13_MultiTurn`	`multi_turn_impossibility: ... → ∀t, ∃z, sys.f t z = sys.τ ∧ sys.D t z = z ∧ ¬(sys.f t (sys.D t z) < sys.τ)`
Theorem 4.3 Stochastic Impossibility	Expectation-preserving stochastic defense fixes a boundary point.	`MoF_13_MultiTurn`	`stochastic_defense_impossibility: ... (∀x, f x < τ → g x = f x) → ... → ∃z, f z = τ ∧ g z = τ ∧ ¬(g z < τ)`
Theorem B.1 Continuous Relaxation	Discrete observations extend to continuous f satisfying Theorem 3.1.	`MoF_Continuous-Relaxation`	`continuous_interpolant_validates_mof: ... (∀i, f (pts i) = obs i) → ... → (∃z, f z = τ) ∧ (∀D, ... → ∃z, f z = τ ∧ D z = z ∧ ¬(f (D z) < τ))`
Theorem C.1 Volume Lower Bound	$\mu(\mathcal{B}_\varepsilon) \geq V_n(\varepsilon/(4L))^n$ for L-Lipschitz f.	`MoF_17_CoareaBound`	`epsilon_band_volume_lower_bound: ... LipschitzWith L f → 0 < L → ε > 0 → f c = τ - ε/2 → μ(Metric.ball c (ε/(4·L))) ≤ μ(coareaBand f`
Theorem C.2 Cone Measure Bound	Closest formal witness: the Euclidean L-based local cone bound gives interval inclusion, positive volume, and post-defense unsafety on that interval.	`MoF_18_ConeBound`	`cone_measure_bound: ... c > L · (K + 1) → (∀t, 0 < t < δ₀ → f (z + t) ≥ τ + c · t) → (∀x ∈ Ioo z (z + δ₀), f (D x) > τ), hence 0 < volume {x | x ∈ Ioo z (z + δ₀) ∧ f (D x) > τ}`
Theorem C.3 Defense Dilemma (K^*)	Closest formal witness: coarse threshold $K^* = G/L - 1$ for persistence vs. utility trade off.	`MoF_19_Optimal-Defense`	`complete_defense_characterization: L > 0 → (G ≤ L → ...) ∧ (G > L → optimalK G L > 0 ∧ (∀K < optimalK G L, G > L · (K + 1)) ∧ (∀K ≥ optimalK G L, L · (K + 1) ≥ G))`
Theorem D.1 Pipeline Lipschitz Degradation	Composition of Lipschitz stages has product constant.	`MoF_15_Nonlinear-Agents`	`lipschitz_comp: LipschitzWith Kf f → LipschitzWith Kg g → LipschitzWith (Kf · Kg) (f ∘ g)`
Theorem D.2 Pipeline Impossibility	Utility-pres. continuous pipeline has boundary fixed points.	`MoF_15_Nonlinear-Agents`	`pipeline_impossibility: ... Continuous P → (∀x, f x < τ → P x = x) → ... → ∃z, f z = τ ∧ P z = z ∧ ¬(f (P z) < τ)`

Table 24. Paper-to-Lean correspondence, continued: auxiliary geometry, attack, stability, and cost results.

Paper result	Informal statement	Lean file	Lean identifier / signature
Theorem E.1 Basin Structure	$\{f > \tau\}$ is open; positive measure under open-positive measures.	`MoF_02_Basin-Structure`	`basin_isOpen` / `basin_measure_pos`: `Continuous f → IsOpen{x | f x > τ}; ... [μ.IsOpenPosMeasure] → (∃p, f p > τ) → 0 < μ{x | f x > τ}`
Theorem E.2 Basin Fragment Min. Size	Connected component of U_τ around p has diameter $\geq 2(f(p) - \tau)/L$.	`MoF_Adv_07_Fragmen-tation`	`basin_fragment_minimum_size`: `LipschitzWith L f → 0 < L → f p > τ → ‖v‖ = 1 → 0 < ε → ∃a b, a,b ∈ superlevelSet f τ ∧ dist a b ≥ 2 · ((f p - τ)/L) - ε`
Theorem H.1 Perturbation Robustness	L-Lipschitz f, $f(p) > \tau$: ball of radius $(f(p) - \tau)/L$ stays unsafe.	`MoF_04_Lipschitz-Basin`	`lipschitz_basin_ball` (with `basin_ball_subset`): `LipschitzWith L f → 0 < L → f p > τ → dist p' p < (f p - τ)/L → f p' > τ`
Theorem H.2 Iterative Convergence	Monotone score-improving operator converges; $\geq \delta$-step finishes in $\lfloor 1/\delta \rfloor$.	`MoF_05_Monotone-Convergence`	`attackSeq_convergent`, `finite_steps_bound`: `∃L, Tendsto (attackSeq op x₀) atTop (nhds L); 0 < δ → (∀x, op.score x + δ ≤ op.score (op.T x)) → n · δ ≤ 1`
Theorem H.3 Transferability	$\|f - g\|_\infty \leq \delta$: $\{f > \tau + \delta\} \subseteq \{g > \tau\}$.	`MoF_06_Transfer-ability`	`transfer_attack` (with `basin_containment`): `(∀x, |f x - g x| ≤ δ) → f p > τ + δ → g p > τ`
Theorem H.4 Authority Monotonicity	Monotone-in-authority f: vulnerability upward-closed; IVT threshold.	`MoF_07_Authority-Monotonicity`	`authority_escalation`, `critical_threshold_exists`: `MonotoneInAuthority f → VulnerableAt f τ y x → x ≤ x' → VulnerableAt f τ y x'; Continuous (f y) → f y x_lo ≤ τ → f y x_hi > τ → ∃x*, f y x* = τ`
Theorem H.5 Gradient Ascent	Nonzero Fréchet derivative ⇒ ascent direction exists.	`MoF_10_Gradient-Attack`	`multivariate_ascent_direction`: `HasFDerivAt f f' x → f' ≠ 0 → ∃v ε, ε > 0 ∧ f (x + ε · v) > f x`

Table 25. Paper-to-Lean correspondence, continued: stability under fine-tuning and cost asymmetry.

Paper result	Informal statement	Lean file	Lean identifier / signature
Theorem I.1 Interior Stability	$\|f-g\|_\infty \le \varepsilon$, $f(x) > \tau + \varepsilon \Rightarrow g(x) > \tau$ (dual for safe).	`MoF_Adv_03_FineTuning`	`small_perturbation_preserves_interior` (and `..._safe_interior`): `(∀x, |f x - g x| ≤ ε) → f x > τ + ε → g x > τ`
Theorem I.2 Crossing Preservation	Transversal crossing with margin $m > \varepsilon$ survives any ε-perturbation.	`MoF_Adv_08_Stability`	`perturbed_crossing_ivt` (or `number_of_crossings_stable`): `Continuous g → (∀x, |f x - g x| ≤ ε) → f a < τ - ε → f b > τ + ε → ∃z' ∈ Icc a b, g z' = τ`
Theorem I.3 Patching Is Nonlocal	$\exists f, g$ with $\|f-g\|_\infty \le \varepsilon$: patching one point shifts distant values.	`MoF_Adv_03_FineTuning`	`patching_nonlocal`: `0 < ε → ∃(f g : ℝ → ℝ), (∀x, |f x - g x| ≤ ε) ∧ ∃τ z, f z = τ ∧ g z > τ`
Theorem J.1 Exponential Cost Asymmetry	Defense cost N^d vs. attack $\le 1/\delta$; ratio diverges in d.	`MoF_Cost_08_CostRatio`	`cost_ratio_tends_to_infinity` (with `attacker_advantage_quantified`): `2 ≤ N → 0 < δ → Tendsto (fun d ⇒ costRatio N d δ) atTop atTop`